

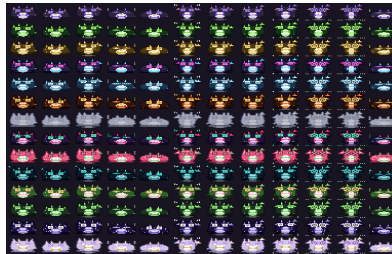
DON'T WATCH MEDIA

TAP AGAINST ME

COMPREHENSIVE GAME BIBLE

Competitive Tap Against Me + Infinity Taps

One shared universe. Two player-facing experiences. Every mechanic, system, content family, intention, and delivery gate consolidated from nine supplied bibles.



Every Tap Against Me Takes You Further.

Canonical compilation | Version 2.0 | July 2026

Document Control

Field	Canonical value
Document	Tap Against Me Comprehensive Game Bible
Version	2.0 - canonical consolidation
Compilation date	July 2026
Feature owner	Don't Watch Media
Primary platform	Android-first, portrait-oriented mobile 2D
Products	Competitive Tap Against Me and Infinity Taps
Authority rule	Newest explicit decision wins; locked/confirmed rules outrank working, candidate, tuning, open, and legacy material.
Source count	Nine supplied files; eight unique documents because two Game Design Bible PDFs are byte-identical.

CANONICAL This document is the consolidated product-intent authority. It is not a substitute for code specifications, balance sheets, legal terms, platform submissions, sprint tickets, or live-operations runbooks. Those artifacts should link back to the relevant chapter and decision status here.

Decision Status Legend

Status	Meaning	Implementation treatment
LOCKED	Core identity, fairness, or non-negotiable boundary.	Do not change without feature-owner approval and a recorded revision.
CONFIRMED	Approved product or system direction.	Implement as written; exact engineering details may still be specified separately.
WORKING	Current intended direction with unresolved implementation or policy detail.	Design modularly and keep tunable values configurable.
CANDIDATE / PROPOSED	Strong option not yet approved as committed scope.	Do not schedule as required launch work without a decision.
TUNING	The behavior is intended; exact values need playtesting.	Expose values through data or remote configuration.
OPEN	Unresolved decision with material product impact.	Record the decision before dependent implementation.
LEGACY / RESERVED	Historical idea retained for context or possible future reuse.	Do not treat as canonical unless revived explicitly.

Implementation Evidence Legend

Design status and implementation status answer different questions. A LOCKED design can still be NOT STARTED; an IMPLEMENTED feature can still embody legacy behavior that the canonical design supersedes.

Evidence state	Meaning	Required next action
IMPLEMENTED	Repository or build evidence shows the feature exists across its stated scope.	Verify against acceptance gates; implementation alone is not release proof.
PARTIAL	A usable subset exists, but required phases, paths, devices, or edge cases remain.	Name the missing scope and the next measurable gate.
DATA-DEFINED	Catalog/configuration records exist, but runtime reachability and grant behavior are not proven here.	Exercise every record through automated and in-build validation.

Evidence state	Meaning	Required next action
PLANNED	The design is documented and sequenced, with no complete implementation evidence in this compilation.	Create the technical specification and milestone evidence.
NOT STARTED	The feature is intentionally absent from the current baseline.	Do not imply current availability; retain dependencies and phase.
UNKNOWN / AUDIT	The source set does not establish current repository or production status.	Perform a code, build, backend, and catalog audit before scheduling or claiming completion.

Conflict Resolution Rules

- The newest explicit decision supersedes older conflicting language.
- Competitive TAM and Infinity Taps are separate player-facing games under one shared TAM ecosystem, even where older documents call Infinity a mode inside one app.
- Core emotional purpose outranks feature quantity: Competitive remains fast, readable, fair, and tense; Infinity remains tactile, alive, relaxing, and non-punitive.
- No pay-to-win is absolute. No purchase may alter competitive physics, visibility, matchmaking, tap validity, survival, item odds, rating, or enforcement.
- Infinity-earned exclusive rewards cannot be purchased in either product. Shared ownership never erases acquisition source.
- Gravity and competitive day/night presentation are independent. Darkness never represents gravity strength in the final implementation.
- Configuration absorbs uncertainty. Gravity curves, spawn rules, bot reactions, matchmaking weights, cycle timing, effect density, and audio compression remain data-driven.
- One deeply reactive Infinity biome is better than many decorative but shallow biomes; one trustworthy ranked queue is better than many fragmented modes.

How to Use This Bible

- Product and design: verify new work against the product architecture, emotional contracts, design pillars, and non-negotiable boundaries.
- Engineering: derive code-level specifications, data models, migrations, tests, performance budgets, and release gates from the relevant chapters.
- Art and animation: use the hero, biome, cosmetic, readability, and reaction standards as the production brief; do not trade competitive clarity for spectacle.
- Audio: preserve a recognizable shared TAM motif while keeping tap, pickup, debuff, match, and accessibility cues legible at all times.
- Live operations: use seasons and events to extend the universe without invalidating permanent ownership or destabilizing ranked rules.
- QA: translate every definition-of-done and appendix acceptance criterion into traceable test cases.

Source Manifest and Provenance

All nine supplied files were inspected. The two files named Game Design Bible and Game Design Bible (1) have the same SHA-256 hash and are exact duplicates. They appear separately below for custody and provenance, but their content is counted once when resolving authority.

Source	Format	Extent	SHA-256 prefix	Contribution
Competitive Dynamic Day-Night and Gravity Presentation Bible	PDF	39 pages	5fa52cf3cf90	Approved competitive environment and gravity-presentation specification.
Game Design Bible (1)	PDF	38 pages	406bba2282e7	Consolidated design, implementation state, and developer acceptance criteria.
TAM Game Design Bible v1.0	DOCX	808 paragraphs	17e44cb45fa0	Newest comprehensive decision register; primary canonical narrative.
Game Design Bible	PDF	38 pages	406bba2282e7	Exact duplicate of Game Design Bible (1); retained in provenance, not double-counted.
TAM Game Bible	DOCX	652 paragraphs	d8aa7d98f6bb	Comprehensive product, content, art, audio, technology, and roadmap synthesis.
TAM Design Bible v1.0	PDF	30 pages	dc5b2cda2b3c	Condensed high-level design and product framing.
Game Bible v1.0	DOCX	643 paragraphs	60f0a151ab57	Detailed mechanics, system behavior, implementation snapshot, and handoff.
Game Design Bible v1.0	PDF	35 pages	889f9f040585	Detailed design, acceptance, roadmap, and production framing.
Original Expanded Bible	PDF	130 pages	8d75129c310f	Deep specifications for effects, Infinity Taps, monetization, and ranked systems.

SOURCE POLICY Chapter source notes identify the principal canonical basis, but many chapters synthesize details from several bibles. The Source Concordance appendix records major cross-source contributions.

Contents

Part	Coverage
Part I	Canon, identity, audience, and shared product architecture
Part II	Competitive TAM mechanics, modes, matchmaking, ranked, and dynamic environment
Part III	Infinity Taps, living worlds, discovery, heroes, and exclusive progression
Part IV	Progression, achievements, cosmetics, effects, economy, and collection
Part V	Music, sound, interface, onboarding, profiles, and social presentation
Part VI	Accounts, backend, security, analytics, accessibility, performance, and mobile quality
Part VII	MVP definitions, phased roadmap, live operations, risks, and decisions
Part VIII	Detailed catalogs, data references, acceptance criteria, terminology, and source concordance

Detailed Contents

All entries use real heading styles for the Word navigation pane and PDF search/bookmark export. This static chapter map avoids field-update drift between editors.

Part	Chapters	Detailed coverage
I	1-6	Governance; executive definition; two-product architecture; vision; audience; universe/brand
II	7-14	Competitive core; controls/physics; modes; pickups/hazards; bots; matchmaking/ranked; ranked implementation; day/night specification
III	15-20	Infinity core; worlds/zones; living interactions; heroes/collection; Velvet Hex/cast production; Infinity implementation
IV	21-25	Progression/economy; achievements/challenges; cosmetics; expanded effect catalog; effect engineering handoff
V	26-27	Music/sound/adaptive audio; interface/onboarding/profiles/social presentation
VI	28-33	Accounts/backend; fair play/analytics; monetization guardrails; economy proposal; accessibility/performance; mobile/session baseline
VII	34-37	MVP/roadmap; production/live ops; risk controls; canonical decisions/open questions
VIII	38-50	Terminology; seed catalogs; acceptance; legacy; feature ledger; implemented catalogs; data/config/events; release gates; decisions; concordance

PART 1

PRODUCT CANON AND GOVERNANCE

The decisions that define what TAM is, who it serves, how the two products relate, and which boundaries cannot be crossed.

1. Document Governance and Decision Status

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

Tap Against Me has evolved quickly through concept design, prototype implementation, backend deployment, multiplayer planning, economy design, audio planning, and the expansion of Infinity Taps. A single design bible is necessary because some ideas were approved, some were refined, and some remain useful but should not silently become requirements.

1 Status Labels

Label	Meaning	How the team should treat it
Confirmed	Explicitly approved or repeatedly established as a core rule.	Implement unless a later product decision replaces it.
Working direction	The current preferred solution, but tuning or implementation details remain open.	Build toward it while preserving configurability.
Future / test candidate	A valuable concept that requires prototyping, balance testing, or later scope.	Do not include in MVP unless it is cheap and supports the core.
Legacy / reserved	An earlier concept, alternate naming, or system that is not part of the current core.	Keep for reference; do not reintroduce without an explicit decision.

2 Conflict Resolution

- The newest explicit decision wins. The current direction is two separate player-facing games under one shared TAM ecosystem, even though earlier descriptions treated Infinity Taps as a mode inside one app.
- Core emotional purpose outranks feature quantity. Competitive TAM must remain fast, readable, fair, and tense. Infinity Taps must remain relaxing, tactile, alive, and non-punitive.
- No pay-to-win is absolute. A monetization idea that affects ranked strength, tap validity, survival, matchmaking, movement, or competitive resources is rejected.
- Infinity exclusivity is meaningful. Rewards designated as Infinity-earned cannot be purchased merely because doing so would be convenient for the shop.
- Configuration should absorb tuning uncertainty. Gravity curves, item spawn rates, bot reaction delay, zone duration, and audio compression should be data-driven rather than hard-coded.

3 What This Bible Is Not

This is not a final production schedule, legal specification, platform-store submission, network protocol reference, or complete balance spreadsheet. It is the master expression of product intent. Engineering specifications, economy tables, art briefs, and sprint plans should derive from it and link back to the relevant design decisions.

2. Executive Definition

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

Tap Against Me is a mobile-first, portrait-oriented 2D tapping universe built around the idea that a simple tap can become competition, movement, expression, discovery, and long-term identity.

CONFIRMED The official slogan is "Every Tap Against Me Takes You Further."

The phrase "me" is intentionally layered. In competitive play, it is the opponent. In solo survival, it is gravity, fatigue, timing, and the player's previous record. In Infinity Taps, it is distance, curiosity, and the limit of what the player has discovered. The title is therefore both a challenge and a promise.

1 The Two Core Experiences

Dimension	Competitive Tap Against Me	Infinity Taps
Primary emotion	Tension, mastery, rivalry, improvement	Calm, curiosity, tactile satisfaction, discovery
Session shape	Short, high-pressure runs and matches	Open-ended, interruption-friendly exploration
Failure	Ground contact or mode-specific loss condition	No traditional failure or punishment
Progress emphasis	Skill rating, rank, wins, survival, seasonal status	Lifetime taps, biome discovery, world interactions, exclusive finds
Population need	Benefits from opponents, bots, ghosts, and matchmaking	Works fully alone and local-first
Content demand	Physics, balance, modes, netcode, competitive integrity	Biomes, animals, reactions, secrets, music, ambient content
Shared systems	Account, heroes, cosmetics, effects, audio identity, achievements	Account, heroes, cosmetics, effects, audio identity, achievements

WORKING Publish Competitive TAM and Infinity Taps as separate player-facing games while sharing as much account, catalog, cloud-save, character, cosmetic, achievement, and backend infrastructure as practical.

WHY SEPARATION MATTERS The two experiences share a verb and a universe, but they sell different emotional promises. A player opening a ranked survival game should not be forced through a relaxation-game onboarding flow, and a player seeking a fidget world should not be confronted with rank anxiety. Separate listings and onboarding allow each product to be clear while the shared ecosystem protects development efficiency.

3. Product Architecture: One Ecosystem, Two Player-Facing Games

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Competitive Product

The competitive product retains the primary name Tap Against Me. It is the sharper, more skill-oriented expression of the brand. It focuses on staying airborne, surviving increasing gravity, managing pickups, competing against human or disclosed AI opponents, and proving improvement through results.

- Solo survival and personal-best play.
- Local play where supported.
- Private online matches between known players.
- Public matchmaking with population-aware bot backfill.
- Ranked ladder with Glicko-style rating and 90-day seasons.
- Permanent progression, achievements, badges, profiles, and cosmetics.

2 Infinity Product

Infinity Taps is the relaxed exploration expression of the universe. A possible store-facing subtitle is "A Tap Against Me Adventure," but final naming remains a brand decision. It should be understandable as a complete product, not as a lesser practice mode.

- No loss condition and no forced session endpoint.
- Tapping moves the hero through responsive spaces.
- Approximately two-minute biome zones create a gentle journey cadence.
- Animals, plants, water, terrain, heroes, and hidden events react to taps.
- Lifetime tap progression and discovery replace ranked pressure.
- Mode-exclusive rewards create status for players who prefer exploration over competition.

3 Shared Ecosystem

CONFIRMED The products should share brand identity, characters, heroes, cosmetics, tap effects, music motifs, account technology, and portions of achievement infrastructure.

WORKING Shared accounts and cloud saves should allow a player's owned heroes, eligible cosmetics, profile identity, and cross-title achievements to appear across both games.

- Shared does not mean identical. Competitive rating, seasonal rank, Infinity discovery progress, and product-specific tutorials remain separate.
- Cross-title ownership must respect source. An Infinity-exclusive skin may be wearable in Competitive TAM after it is earned, but it cannot be purchased in the competitive shop.
- Storefront clarity comes first. Each listing must clearly describe its own loop rather than relying on the player to understand the broader universe.
- Technical reuse should be invisible. Players should feel continuity, not a stitched-together launcher.

4 Recommended Account Boundary

Data	Shared	Product-specific
Identity	User ID, display name, profile frame, primary badge	Ranked banner treatment or Infinity exploration title
Collection	Owned heroes and globally eligible cosmetics	Product-restricted equipment slots or encounter records
Currency	Hero Coin wallet, subject to economy safeguards	Event currencies when necessary
Achievements	Universal and cross-title achievements	Ranked, survival, biome, and discovery achievements
Progress	Lifetime total taps may be aggregated with source labels	Glicko rating, season rank, biome journal, world distance
Settings	Accessibility and account preferences	Control and audio profiles tuned per title

4. Vision, Identity, and Design Pillars

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Vision Statement

Build the most expressive tapping universe on mobile: easy to understand in seconds, satisfying to touch repeatedly, deep enough to reward mastery or curiosity, and fair enough that achievement remains meaningful.

2 Shared Design Pillars

Immediate tactile feedback

Every tap should produce a readable response through motion, sound, particles, environmental reaction, or character behavior. Dead taps should be rare and intentional.

Simple input, layered mastery

The basic action is tapping. Depth comes from timing, location, momentum, survival decisions, exploration, pattern recognition, and collection-not from a crowded control scheme.

Personality through reaction

Heroes and environments should react as if they notice the player. Repeated taps should escalate behavior instead of replaying one animation forever.

Fair status

Ranks, badges, exclusive rewards, and rare discoveries matter only when the method of earning them is trustworthy.

Respect for session context

Competitive TAM supports focused short sessions. Infinity Taps supports fidgeting, commuting, listening to music, or briefly returning throughout the day.

Expandable universe

Characters, variants, biomes, effects, achievements, seasons, and music can grow without rewriting the core verb.

3 Competitive-Specific Pillars

- A loss should feel understandable rather than arbitrary.
- The player should believe improvement is possible after every run.
- Matchmaking should create pressure without creating hopeless pairings.
- Bots must behave imperfectly and remain bound by the same physical rules.
- Rank must reflect sustained performance, not a single lucky run or paid advantage.

4 Infinity-Specific Pillars

- The world should feel alive before the player touches it.
- Taps influence the world; they do not merely trigger cosmetic overlays.
- Nothing in the environment should punish the player with failure.
- Discovery rewards observation, repetition, experimentation, and patience.
- Exclusive Infinity rewards validate noncompetitive play as a first-class way to engage with TAM.

5. Audience and Emotional Experience

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Competitive Audience

Competitive TAM is for players who enjoy short repeatable challenges, personal-best chasing, synchronous or asynchronous rivalry, visible rank, collectible identity, and controls that become more expressive with practice. The ideal player can understand the premise immediately and still care about improving weeks later.

2 Infinity Audience

Infinity Taps is for players who tap to fidget, regulate attention, relax, fill short pauses, watch satisfying reactions, collect unusual variants, or explore without the threat of losing progress. It should remain engaging when played with sound low, when interrupted, or when the player has no desire to compete.

3 Dual-Audience Player

Many players will move between both products. A ranked player may use Infinity Taps to decompress. An exploration player may eventually try solo survival because a familiar hero and earned cosmetic carry over. Cross-title design should encourage curiosity without using coercive rewards that make either game feel mandatory.

4 Emotional Contract

Player expectation	Competitive promise	Infinity promise
My tap matters	It changes velocity, survival, positioning, or match outcome.	It causes visible, audible, or systemic world response.
My time matters	Results, rating, records, and rewards are recorded fairly.	Lifetime taps, discoveries, encounters, and exclusive unlocks persist.
My collection matters	Cosmetics show identity without changing power.	Discoveries and exclusives show curiosity and exploration history.
The game respects me	No hidden paid advantage or impossible matchmaking.	No surprise failure, pressure, or punishment for pausing.

6. Shared TAM Universe and Brand Language

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Naming

- Tap Against Me (TAM): the competitive flagship and umbrella universe name.
- Infinity Taps: the no-loss exploration title. "A Tap Against Me Adventure" is a possible clarifying subtitle, not yet final.
- Hero: the current canonical term for playable or collectible characters.
- Hero Variant: a visually or thematically distinct version of a hero with separate ownership or unlock status.
- Hero Coin: the canonical cosmetic currency. Earlier "Golden Hero Token" language can remain as the physical in-match pickup representation.

2 Meaning of "Me"

The identity should remain slightly mysterious rather than over-explained in marketing. "Me" can be the opponent, gravity, the game, a previous score, or the part of the player that would stop. This interpretation makes the slogan equally valid in ranked competition and peaceful exploration.

3 Tone

TAM should feel playful, energetic, slightly mischievous, and sincere about player achievement. Humor should come from character reactions, rare events, environmental surprises, and over-the-top cosmetic consequences-not from mocking the player for losing.

4 Visual Language

- Readable silhouettes suited to small portrait screens.
- Bright effects layered over clear gameplay boundaries.
- Purple as a recurring identity color, with gold tied to Hero Coins and prestige.
- Cute or strange heroes capable of expressive multi-stage reactions.
- Environments that feel hand-authored even when assembled procedurally.
- Pixel-art or stylized 2D presentation may be used, but readability is more important than strict aesthetic purity.

5 Character Lore

Character lore should be compact, collectible, and discoverable. A hero can have a name, one-sentence identity, a few journal notes, variant-specific flavor, and environmental behavior. Repeated encounters can reveal personality gradually.

LEGACY Earlier work referenced "Puffs," artifacts, and bubble-like moving targets. These concepts may inform future companions, collectible lore objects, or event modes, but heroes are the current primary character language.

PART 2

COMPETITIVE TAP AGAINST ME

The skill-focused survival game: input, physics, pickups, modes, opponents, rating, seasons, presentation, and competitive trust.

7. Competitive Tap Against Me: Core Gameplay

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Core Loop

1. The player enters a run or match and begins falling from the sky or from an airborne starting state.
2. Each valid tap applies upward force, allowing the player to remain airborne.
3. Gravity continually pulls the hero downward and increases over time or by difficulty state.
4. The player responds to movement, pickups, arena conditions, and an opponent where applicable.
5. Touching the ground ends the run or eliminates the player under the active mode rules.
6. The result feeds records, achievements, rating, seasonal progress, challenges, and cosmetic progression.

CONFIRMED The fundamental skill test is staying airborne through tapping. Ground contact ends the standard survival round.

2 Why the Loop Works

The loop creates immediate comprehension and escalating tension. Early taps establish rhythm. Mid-run play adds pickup decisions and position control. Late-run play increases physical and cognitive pressure because gravity, speed, or hazards reduce recovery time. The round produces a natural story without requiring complex rules.

3 Player Skill Expression

- Tap timing and rhythm.
- Recovery after a poor tap.
- Positioning relative to pickups and arena boundaries.
- Recognizing when a boost is worth pursuing.
- Managing increased gravity and momentum.
- Reading the opponent without losing control of one's own hero.
- Maintaining consistency as the run becomes faster and more stressful.

4 Standard Failure Readability

A player should know why they lost. The final seconds should be reviewable through clear motion, a brief result recap, and optionally a simple trace of the last taps or velocity state. Sudden hidden modifiers, ambiguous ground collision, or network corrections that appear as unexplained deaths undermine trust.

5 Session Targets

Competitive sessions should support rapid rematches and short attention windows. Survival averages will vary by skill; the matchmaking discussion uses examples such as a 15-second player versus a 72-second player to illustrate the need for meaningful skill segmentation. The game should not assume one universal target duration.

8. Competitive Controls, Physics, and Difficulty

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Input

The primary input is a screen tap. The game must feel responsive without allowing input noise, accidental edge taps, or device sampling differences to create unfair advantages.

WORKING Tap position may influence horizontal direction or movement bias, especially where left/right navigation is needed. This should remain configurable and be validated against one-thumb accessibility.

2 Physics Principles

- Upward impulse: a valid tap adds upward velocity or force rather than teleporting the hero.
- Momentum: movement should carry naturally enough that recovery and overcorrection matter.
- Gravity curve: gravity can intensify over time to create escalation, but the curve must remain learnable.
- Ceiling behavior: the game should avoid allowing a player to park safely at the top. A ceiling response, diminishing lift, or arena rule should keep movement active.
- Collision clarity: the ground boundary must be visually obvious and collision should match the art.
- Frame independence: physics and tap validation must be deterministic or normalized across supported refresh rates.

3 Difficulty Escalation

CONFIRMED Gravity increases over time in the core survival concept.

CANDIDATE Additional escalation tools include stronger wind, fewer beneficial pickups, more poisonous fruit, moving or shrinking safe regions, environmental hazards, and faster arena motion.

Difficulty should escalate through one or two readable variables at a time. Layering every hazard simultaneously creates noise rather than mastery. Ranked modes require fixed and published rule sets; experimental combinations can live in events or unranked modes.

4 Tap Validation

- Count only taps received during active gameplay and inside valid input regions.
- Prevent menus, ads, notifications, or overlays from generating gameplay taps.
- Record tap timestamps for local analysis and server verification where appropriate.
- Use reasonable upper bounds and behavioral analysis rather than punishing fast legitimate players solely for high taps per second.
- Separate display tap count from authoritative match simulation when network play requires reconciliation.

5 Feel Tuning

The best physics are not necessarily the most realistic. The hero should feel bouncy, expressive, and recoverable early, then increasingly demanding. Hit-stop, squash-and-stretch, subtle camera response, and tap sounds can increase perceived responsiveness without changing simulation power.

9. Competitive Modes and Match Structures

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Launch-Critical Modes

Mode	Purpose	Population requirement	Launch priority
Solo Survival	Personal best, practice, achievement progress, baseline performance profile	None	Essential
Local Play	Immediate social play and testing where device format permits	Local participants	Useful
Private Online Match	Direct account-to-account validation and friend competition	Known opponent	Essential networking milestone
Public Matchmaking	Core scalable competitive experience	Human pool plus bot backfill	High
Ranked	Seasonal status and long-term mastery	Stable rules, anti-cheat, enough match confidence	After reliable public play

2 Match Rule Families

CONFIRMED The design has discussed Survival, Most Taps, First to Zero, and Dual Victory as possible structures.

Survival

Last airborne wins. This is the cleanest expression of the core mechanic and should remain the primary competitive rule until alternate modes prove equally readable.

Most Taps

Players compete for the most valid taps within a time or survival framework. This mode requires careful anti-macro validation and should not reward meaningless tapping that ignores movement skill.

First to Zero

A life-based mode where one player reduces the other to zero. The exact damage and regeneration model has not been finalized. Earlier concepts included life, regeneration per tap, total life, and tap-rate upgrades; these must never become purchasable competitive power.

Dual Victory

A match can recognize multiple accomplishments-for example, outlasting an opponent while also meeting a tap or objective condition. This should be presented as a bonus reward or event rule rather than making standard victories difficult to understand.

Tournament and Free-for-All

CANDIDATE Real-time free-for-all, elimination brackets, tournaments, and survival leaderboards are valid expansion directions after 1v1 reliability and population are established.

3 Rematch and Result Flow

- Show winner, survival time, valid taps, key pickups, rating change, challenge progress, and earned rewards.

- Provide a fast rematch path.
- Do not bury the reason for a rating change.
- Allow a player to inspect an opponent profile without interrupting rematch momentum.
- Use celebratory cosmetics without obscuring the result.

10. Pickups, Hazards, and Arena Dynamics

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Green Apples

CONFIRMED Green Apples provide a beneficial jump-force or lift boost.

The boost should be strong enough to influence routing but not so powerful that one spawn decides a match. Spawn visibility and pickup timing must allow both players to understand the opportunity.

2 Purple Fruit / Poisonous Fruit

CONFIRMED Purple poisonous fruit increases gravity or otherwise debuffs airborne control.

Poison fruit can create risk, denial, or achievement opportunities. The player should see when poison is active and understand its duration or severity. Achievements such as Life of Poisonous Fruit, Toxic Appetite, Poison-Proof, Maximum Poison, and Bad Decisions support the identity of this system.

3 Hero Coin Pickup

CONFIRMED Golden Hero tokens were discussed as spawning pickups spendable in the shop. The canonical currency name should be Hero Coin; a golden token can be its visual pickup form.

Competitive coin rewards must be tuned so players do not intentionally sabotage matches to farm. Ranked rewards, completion bonuses, and capped pickup value are safer than unlimited spawning.

4 Pickup Responsiveness

CONFIRMED Current pickup behavior has felt too reactive and quick. Item collection or bot response to items should be slowed and made more human-readable.

- Add a brief anticipation window or proximity threshold.
- Avoid bots snapping to a pickup on the exact frame it appears.
- Give pickups readable spawn animation and stable collision.
- Allow high-skill interception without making every item deterministic.
- Log pickup timestamps and contest outcomes for balance analysis.

5 Arena Dynamics

CANDIDATE Wind, moving ground, shrinking regions, moving obstacles, environmental hazards, and themed arenas can expand variety.

Arena modifiers should be classified as ranked-safe, unranked, or event-only. Ranked-safe modifiers must be symmetrical, clearly telegraphed, and deterministic. Cosmetic arena themes must not hide collision edges or pickups.

11. Bots, Ghosts, and Population Strategy

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Why Bots Are Needed

A new competitive game cannot assume a healthy matchmaking pool at every rating, region, and hour. Bots prevent empty queues and allow immediate testing. They are a population bridge, not a substitute for authentic competition.

CONFIRMED Bots are an acceptable solution for public matches when human opponents are unavailable, provided they do not behave like perfect tapping machines.

2 Bot Fairness Rules

- Bots use the same gravity, tap impulses, momentum, collision, pickups, character stats, abilities, and arena rules as players.
- Bots do not read future random events or hidden state.
- Bots have reaction delay, attention limits, execution error, and recovery behavior appropriate to their rating.
- Bots do not instantly acquire items or perfectly optimize every tap.
- Bots should be disclosed as CPU or AI challengers rather than disguised as human players.
- Bot difficulty is derived from the player's matchmaking band and recent performance, not from predatory loss manipulation.

3 Bot Behavior Model

A credible bot should operate from imperfect perception. It estimates vertical velocity, distance to ground, available recovery, nearby pickups, and opponent state. It chooses among survival, pursuit, pickup, denial, and conservative recovery behaviors. Noise should affect timing and target choice, not merely add random failure.

4 Ghost Opponents

WORKING Recorded-run ghost opponents may be an even stronger early population solution for asynchronous competition.

A ghost replays a real player's timestamped input or authoritative movement trace. Ghosts provide human rhythm and mistakes without requiring simultaneous presence. They can power daily challenges, rival replays, friend challenges, and rank-adjacent modes. They should be labeled as recorded runs and protected against stale physics versions.

5 Ranked Restrictions

WORKING Lower and mid tiers may permit disclosed bot backfill when queues are thin. High tiers should increasingly prioritize or require human-versus-human matches.

The exact cutoff should be based on population data. A player's ranked badge must not imply human-only achievement if a substantial portion of the path was against undisclosed or exploitable bots.

12. Matchmaking, Glicko Rating, Tiers, and Seasons

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Rating Model

CONFIRMED The preferred rating approach is Glicko-style rather than simple Elo, combined with a rolling survival-performance profile.

Glicko is useful because it tracks both rating and uncertainty. New or inactive players can move more quickly until the system becomes confident. Rating alone is not enough for TAM because two players may have similar win records while having dramatically different survival capacity.

2 Performance Profile

Matchmaking should maintain recent and lifetime metrics such as median survival time, trimmed mean survival time, late-run consistency, valid tap rate, pickup behavior, opponent strength, and result variance. The explicit design example is that someone averaging about 15 seconds should not routinely face someone averaging about 72 seconds. A 63-second player facing a 72-second player may be a productive stretch match.

3 Match Search Expansion

1. Begin with a narrow Glicko and survival-profile range.
2. Expand gradually based on queue time.
3. Prefer a disclosed bot or ghost over an obviously hopeless human mismatch.
4. Protect new players from high-confidence veterans even when surface rating is similar.
5. Use connection quality and region after skill compatibility, not instead of it.

4 Tiers

CONFIRMED The ranked ladder should use visible tiers such as Bronze, Silver, Gold, and Platinum, with additional upper tiers as the system matures.

WORKING The presentation can take inspiration from Hearthstone's approachable tier progression and seasonal climb without copying its exact rules.

- Tiers should communicate broad achievement; Glicko remains the precise hidden or secondary measure.
- Divisions within tiers can create steady milestones.
- Promotion moments should be celebratory but not excessively slow.
- Rank floors may be used carefully at lower tiers to reduce frustration.
- Top-tier placement should depend on credible human competition and stable anti-cheat.

5 Season Structure

CONFIRMED Ranked seasons reset every 90 days.

A reset should not erase permanent collection, achievements, lifetime statistics, or Infinity progress. It should compress or partially reset ranked standing so players have a reason to climb again. End-of-season rewards should be cosmetic and scaled by achieved rank.

6 Seasonal Rewards

- Rank badge or profile frame reflecting the highest achieved tier.
- Season-exclusive hero variant, tap effect, particle, entrance, or victory pose.
- Hero Coins within a controlled range.
- Permanent season record on the profile.
- No competitive stat advantage in the next season.

7 Rating Integrity

- Use authoritative match records and idempotent result submission.
- Detect repeated opponent farming, intentional disconnect patterns, and impossible tap traces.
- Do not punish one disconnect as cheating; use confidence and repeated behavior.
- Version rating calculations and preserve auditability.
- Explain penalties and rating changes in player-readable terms.

13. Competitive Rating, Matchmaking, Bot, and Season Implementation Reference

Source basis: Original Expanded Bible pp. 105-130; Game Bible v1.0 competitive sections; dedicated ranked guidance across consolidated bibles.

CONFIRMED Use Glicko-2 as the underlying skill model and a readable tier/division ladder as the player-facing progression layer. Survival performance protects match quality but does not replace rating.

Rating Record

Field	Initial / canonical direction	Status
Rating	1500	WORKING initial value
Rating deviation	350	WORKING initial value
Volatility	0.06	WORKING initial value
Placement matches	5	CONFIRMED starting count
Season duration	90 days	CONFIRMED
Reset formula	$1500 + ((\text{old rating} - 1500) \times 0.60)$	WORKING; compression configurable

Persist rating, deviation, volatility, visible tier, division, stars/progress, placement state, season ID, peak/final rank, human/bot split, wins/losses, streaks, and match history as separate fields. Never infer matchmaking solely from the visible badge.

Matchmaking Signals

Signal	Initial weight	Purpose
Glicko rating and uncertainty	70%	Primary skill and confidence signal.
Rolling survival profile	20%	Domain safeguard against obviously hopeless pairings while rating is uncertain.
Connection and recent constraints	10%	Latency, region, disconnect behavior, version, and practical match quality.

TUNING Weights are a configurable starting model, not immutable mathematics. Target a modeled 40-60% win-probability range when population allows.

Survival Compatibility Bands

Player profile	Starting opponent window	Interpretation
0-15 seconds	Approximately +/-8 seconds	Protect true beginners and very short survivors.
16-30 seconds	Approximately +/-12 seconds	Developing control.
31-50 seconds	Approximately +/-15 seconds	Intermediate consistency.
51-75 seconds	Approximately +/-18 seconds	Strong players can face credible stretch opponents.
76+ seconds	Rating, percentile, and high-skill population model	Do not force rigid time brackets at the top end.

A 15-second player should not normally face a 72-second player. A 63-second player may reasonably face a 72-second player because the result remains credible and instructional.

Queue Expansion

- 0-5 seconds: narrow rating, survival, region, version, and connection range; prioritize humans.
- 5-10 seconds: widen rating and survival slightly while preserving connection quality.
- 10-15 seconds: widen further inside protected fairness limits and continue prioritizing humans.
- After roughly 15 seconds: insert an appropriate bot where rank and mode policy allow; ranked queues may wait longer.

Bot Reaction and Decision Model

Bot band	Recognition / decision delay	Useful-item pursuit
Beginner	600-1,000 ms	25-45%
Developing	450-800 ms	Tunable between beginner and intermediate
Intermediate	300-650 ms	45-70%
Skilled	200-500 ms	65-85%
Advanced	140-350 ms	Tunable between skilled and elite
Elite	100-280 ms	75-92%

- Bots use the same gravity, impulses, momentum, collisions, pickup rules, arena rules, information, and legal abilities as humans.
- Bots cannot know future spawns, react on the exact spawn frame, recover perfectly, or use hidden movement advantages.
- Human-like errors include early/late taps, overcorrection, panic, ignored items, risky pursuit, poor item use, and failed recovery.
- Performance varies around a target distribution. A 63-second target bot may commonly finish 52-72 seconds, sometimes fail at 40-50, and occasionally reach 73-82.
- Personalities include Floater, Aggressor, Climber, Technician, Trickster, and Panicker; personality changes decisions, never physics.

Visible Ladder and Unresolved Division Count

WORKING Current working default: Bronze through Diamond use divisions V-I; Master is rating-based; Grandmaster is leaderboard-based. A duplicate consolidated PDF instead uses III-I, so division count remains in the Decision Register until formally locked.

Tier	Working presentation	Bot policy direction
Bronze	V-I; 3 stars per division	Allowed
Silver	V-I; 4 stars per division	Allowed
Gold	V-I; 5 stars per division	Reduced frequency
Platinum	V-I; 5 stars per division	Rare; human preferred
Diamond	V-I; 5 stars per division	Emergency only; reduced prestige impact
Master	Rating / leaderboard presentation	Human only
Grandmaster	Leaderboard placement	Human only

Rank floors are recommended at Bronze V, Silver V, Gold V, Platinum V, Diamond V, and Master. Visible win streaks may begin after three wins and grant an example two stars per win in lower/middle tiers; Glicko math never doubles. Streak bonuses stop near Diamond or Master.

Bot Choice and Ranked Reward Policy

An unranked preference may expose Allow Bot Opponents: On / Off with an explicit warning that Off can lengthen queues. Ranked may override the preference under a published tier/queue policy; it must never hide the fact that an opponent is a bot.

Rank band	Bot availability	Candidate rating / visible progression treatment
Bronze / Silver	Allowed	Normal stars; normal or slightly reduced Glicko gain; reduced loss when losing to a bot.
Gold / Platinum	Reduced / rare	Reduced Glicko gains; visible progression remains readable; bot frequency capped.
Diamond	Emergency backfill after extended queue	Reduced gain and loss; repeated bot wins cannot drive leaderboard-critical advancement.
Master / Grandmaster	Human only	No bot backfill; prestige and leaderboard position rely on verified human matches.

Candidate Seasonal Reset Placement

Previous finish	Working new-season start
Bronze	Bronze
Silver	Bronze or Silver V
Gold	Silver
Platinum	Gold
Diamond	Platinum
Master	Diamond
Grandmaster	Diamond or Master placement

TUNING These visible placements are examples layered on the soft Glicko compression. They require simulation and depend on the unresolved III-I versus V-I division decision.

Candidate Peak-Rank Reward Ladder

Highest tier reached	Proposed permanent reward category
Bronze	Hero Coins and seasonal participation badge.
Silver	Additional Hero Coins and profile badge.
Gold	Seasonal tap effect.
Platinum	Animated badge or profile frame.
Diamond	Exclusive character particle effect.
Master	Exclusive seasonal hero/character variant.
Grandmaster	Numbered leaderboard badge and premium cosmetic.

Archive season number, highest rank, final rank, peak Glicko rating, win/loss record, best survival, leaderboard placement, human/bot split, and every granted reward. Reward qualification should use highest rank reached

unless formally revised; final rank remains visible separately.

Authoritative Ranked Result Processing Order

1. Validate match evidence and pinned ruleset/version.
2. Confirm disconnect, abandonment, timeout, server-failure, and reconnect state.
3. Record both survival durations and supporting performance evidence.
4. Record winner, loser, placement, and exact result reason.
5. Update rolling performance statistics.
6. Calculate Glicko-2 rating, deviation, and volatility updates server-side.
7. Update visible stars, points, or division progress without altering Glicko math.
8. Apply win-streak changes and eligible visible bonus.
9. Update seasonal statistics and peak/final tracking.
10. Update bot statistics and anti-farming counters where applicable.
11. Check promotion, demotion, floor, placement completion, and upper-tier eligibility.
12. Grant milestone/seasonal rewards once through ledger/idempotency keys.
13. Save immutable match history and validation outcome.
14. Update eligible leaderboards after finalization.

Disconnect Evidence

Log player disconnect timestamp, reconnect attempts, match state/phase, network error, client closure, server error, grace-window outcome, abandonment history, and final result reason. One unstable connection is not cheating; enforcement uses repeated high-confidence patterns and provides an appeal path.

Leaderboard Families

- Highest verified Glicko rating and Grandmaster placement.
- Highest average survival and longest individual verified survival.
- Most ranked wins and highest win streak.
- Most improved rating during the season.
- Friends-only and, after population/privacy validation, regional views.

Primary prestige boards use verified ranked matches. Bot-heavy statistics are excluded or clearly separated; Master and Grandmaster prioritize human-versus-human evidence.

Ranked Remote-Configuration Registry

Remotely configurable values include season dates, tier thresholds, divisions/stars, floors, placement count, search duration, expansion intervals, survival compatibility, bot insertion timing, reaction ranges, awareness/pursuit, bot availability by tier, rewards against bots, rating compression, streak requirement/bonus, seasonal reward table, disconnect grace, and queue-penalty duration. Every match and season pins an immutable config version for replay and audit.

Ranked MVP and Post-MVP Boundary

Scope	Included content
Initial MVP	Glicko-2; Bronze-Diamond; division/stars/floors after decision; 5 placements; rolling and recent-10 survival; rating+survival matchmaking; expanding search; bot backfill; 3 bot bands; randomized reactions/item delay; 90-day reset; rewards; history; server result validation; basic anti-farming.
Post-MVP	Full bot personalities; advanced performance profile; friend/regional/Grandmaster leaderboards; ghosts; replay; spectating; seasonal stats pages; placement prediction; smurf/disconnect analysis; dynamic bot calibration; automated matchmaking tuning; richer reward tracks; tournaments; inactivity decay for upper tiers.

Ranked Definition of Done

- Two real players complete a ranked match and both Glicko records update correctly on the server.
- Rating uncertainty changes update magnitude; visible stars, floors, promotions, and demotions behave as published.
- Five placements produce a credible visible starting rank.
- Queue widening is gradual, logged, and never normally pairs a 15-second player with a 72-second player.
- Bots are visibly identified, fallible, rank-appropriate, and protected against repeat farming.
- The 90-day reset soft-compresses rating, increases uncertainty, preserves history, and grants the correct peak-rank reward.
- Disconnect, abandonment, verified server failure, duplicate submission, and reconnect rules produce one explainable result.

14. Competitive Dynamic Day-Night and Gravity Presentation

Source basis: Competitive Dynamic Day-Night and Gravity Presentation Bible v1.0, pp. 1-39. Approved design direction; implementation values remain subject to playtesting and performance validation.

LOCKED Time changes atmosphere. Gravity changes movement. The two systems may advance from the same authoritative match clock, but neither calculates, controls, or explains the other.

Feature Intention

The competitive arena should feel like a world moving through time rather than static wallpaper. Sunrise, daylight, sunset, twilight, stars, moonlight, nocturnal ambience, and predawn provide atmosphere and a subtle sense of endurance. Mechanical difficulty remains in the gravity system. The environment remains synchronized, cosmetic, readable, and competitively neutral.

- Make the competitive world feel alive without competing for attention.
- Give long survival runs a visual sense of accomplishment without replacing the timer or score.
- Increase match-to-match variety through seed-controlled starting phases.
- Protect characters, opponents, pickups, hazards, effects, warnings, and interface information during every phase.
- Remove the misleading implication that darkness causes gravity or nighttime is punishment.

Locked Cycle Rules

- A complete competitive day/night cycle lasts exactly 120 seconds.
- Transitions are continuous; the game must not switch abruptly among four background images.
- The sun and moon visibly travel through portrait-safe arcs and have no collision or gameplay effect.
- Both synchronized competitors, bots, replays, and spectators use the same authoritative phase.
- Day/night never changes gravity, tap force, movement, collision, spawn probability, bot behavior, scoring, rating, or latency compensation.
- The interface does not inherit world lighting.
- Night cannot materially reduce visibility, and premium cosmetics cannot gain a visibility advantage.
- The 120-to-0 wrap has no flash, palette reset, sprite teleport, asset hitch, or lighting discontinuity.

Authoring Phase Baseline

Elapsed time	Normalized range	Phase	Presentation emphasis
0-12 s	0.00-0.10	Sunrise	Horizon glow; violet to warm orange; stars fade; morning mist/birds may appear.
12-42 s	0.10-0.35	Morning / daylight	Cleanest competitive visibility; sun rises; short shadows; daylight creatures and highlights.
42-60 s	0.35-0.50	Afternoon / sunset	Warmer orange, rose, red, and purple; longer shadows; protect gold/red/yellow pickups.
60-72 s	0.50-0.60	Twilight	Sun below horizon; moon and stars emerge; artificial lights begin; rim-light support increases.
72-108 s	0.60-0.90	Night	Moon arc; full stars; nocturnal ambience; readable characters and interactions; moonlight reflections.
108-120 s	0.90-1.00	Predawn	Moon approaches horizon; stars fade; cool/pale glow reconnects seamlessly to sunrise.

TUNING Phase ranges are authoring guides, not hard scene cuts. Every color, light, shadow, star, ambience, and reaction property should interpolate across boundaries.

Starting-Phase Strategy

The initial implementation begins at fixed sunrise (phase 0.00) so timing, synchronization, contrast, and loop bugs are reproducible. After validation, the server or authoritative match creator selects one curated anchor from the match seed.

Anchor	Normalized phase	Reason for curation
Sunrise	0.00	Clear cycle start and repeatable validation.
Midday	0.25	Neutral high-readability scene.
Sunset	0.45	Warm-color readability test.
Early night	0.62	Night transition and artificial-light scene.
Midnight	0.75	Full night presentation.
Predawn	0.92	Cool low-light return toward sunrise.

A small deterministic offset may later reduce repetition. Completely uncontrolled random starts are discouraged because they create awkward transitions and inconsistent testing.

Celestial and Layering Rules

- Sun and moon travel between opposite horizons on biome-configurable arcs; exact astronomy is less important than clear movement and composition.
- Avoid placing celestial bodies behind the densest gameplay region for extended periods.
- Recommended draw order: sky gradient, stars, sun/moon, distant clouds, far background, mid-background, near background, gameplay, foreground effects, interface.
- No celestial collision, tap response, pickup creation, physics, rarity change, lens flare over gameplay, flash, or blinding effect exists in standard competitive play.

Protected Gameplay Readability

Biome lighting may change dramatically, but the gameplay layer follows protected rules rather than simply brightening everything and flattening the hierarchy.

- Minimum hero and opponent brightness with subtle outlines and nighttime rim lighting.
- Fixed or limited-range pickup emissive values and background dimming behind critical objects when necessary.
- Controlled saturation and particle brightness directly behind gameplay objects.
- Golden Hero Tokens remain distinct during sunset; Green Apples remain distinct from foliage; Purple Fruit remains distinct at twilight and night.
- The timer, names, status, gravity meter, item indicators, connection warnings, ranking, pause controls, prompts, and results remain independent from world lighting.
- Important objects use shape, iconography, outline, animation, contrast, and color rather than color alone.

Gravity Presentation After Decoupling

Removing sky darkness as the gravity indicator requires a richer physical language. Increasing gravity may add faster downward acceleration, stronger downward particle drift, heavier squash, forceful recovery, limited threshold camera impulse, faster debris, descent streaks, downward distortion, environmental sway, impact anticipation, a subtle extreme-pressure vignette, a more active gravity meter, and denser low-frequency audio.

Low-gravity states use longer hang time, slower particle descent, floating accessories, upward dust/pollen, extended arcs, soft impacts, reduced squash, and wider trailing motion.

LOCKED Purple Fruit presentation is player-local and distinct from the time-based gravity ramp: a purple pulse, downward streak, gravity icon/timer, short low-frequency impact, and stronger downward-motion feedback are valid. It must never recolor the global sky or affect the opponent's environment.

Authoritative Environment Data

Field	Purpose
match_id	Connects environment state to the authoritative match.
environment_seed	Reproduces starting phase, decorative variants, clouds, stars, and seeded events.
day_night_profile_id	Selects biome gradients, curves, celestial arcs, ambience, and fallback settings.
cycle_duration	120.0 seconds for standard competitive play.
starting_phase	Curated normalized phase selected by the authoritative match creator.
authoritative_match_start_time	Source of elapsed match time and phase reconstruction.
biome_id	Selects the competitive biome.
environment_variant_id	Selects an allowed seeded visual variant.

Clients calculate phase as $\text{starting_phase} + \text{elapsed} / \text{cycle_duration}$ modulo 1. The server does not send colors every frame. Under 50 ms drift may be ignored; 50-250 ms is smoothed; larger drift is corrected rapidly. Reconnect restores the authoritative phase with a short blend where possible.

Controller Responsibilities

Component	Owns	Must not own
MatchEnvironmentController	Authoritative configuration, phase, synchronization, biome profile coordination.	Gravity calculation.
DayNightProfile	Gradients, curves, celestial arcs, ambience, performance fallbacks.	Player-specific or match-result state.
CelestialController	Sun/moon position, opacity, glow, horizon hiding, safe composition.	Collision or gameplay effects.
SkyRenderer	Gradient, stars, clouds, background-only lighting and low-cost modes.	Gameplay contrast or physics.
GameplayReadabilityController	Hero/pickup contrast, outlines, rim light, minimum brightness, accessibility response.	Collision or simulation.
GravityPresentationController	Particles, squash, pressure, audio, Purple Fruit presentation from effective gravity.	Sky phase.
BiomeReactionController	Seeded day/night creatures, lights, plants, reflections, atmosphere.	Gameplay spawns or hazards.
EnvironmentalAudioController	Crossfaded ambience below gameplay audio.	Restarting or controlling competitive music.

Performance and Accessibility

Tier	Preserved identity	Reduced work
High	Full gradient, celestial movement, readability, ambience.	Uses glow, more stars, dynamic shadows, reflections, particles.
Medium	Full gradient, sun/moon, readability, phase identity.	Reduced stars, shadows, particles, and simplified reflections.
Low	Gradient/ramp sky, sun/moon, minimal stars, protected readability.	Static/disabled dynamic shadows, minimal particles, no expensive reflections.

- Reduced motion: simplify clouds, parallax, animals, gravity streaks, particles, and camera impulses while color progression continues.
- Reduced effects: trim glow, stars, reflections, fireflies, fog, and shadow animation.
- Night visibility: increase outline, rim light, pickup outline, and background separation without revealing hidden information.
- Photosensitivity: no rapid celestial flashes, cycle lightning, star flicker, full-screen pulse, or abrupt exposure shift.

Eight Delivery Phases

0 - Audit and decoupling

Find every gravity-derived tint/shader/global, document existing curves, create separate environment and gravity values, and prove neither reads the other.

1 - Core cycle MVP

One test biome; fixed sunrise; normalized timer; gradient; sun, moon, stars; exact seamless 120-second wrap; stable UI; target Android test.

2 - Readability and art

Six phase palettes, rim light, pickup protection, token/apple/fruit tests, star/cloud/shadow tuning, and neutral editor lighting.

3 - Online synchronization

Environment seed, authoritative phase/time, drift correction, reconnect, private match, bot, replay, and spectator validation.

4 - Starting-phase variety

Curated anchors from the match seed, phase-distribution analytics, readability validation, and seamless transitions from every anchor.

5 - Biome reactions and ambience

Profile resources, day/night reactions, artificial lights, reflections, crossfades, deterministic events, and density limits.

6 - Gravity presentation

Normalized strength, downward particles, squash, thresholds, Purple Fruit feedback, gravity audio, and accessibility controls.

7 - Optimization and release

GPU/shader/overdraw, Android tiers, battery/thermal, 30/60 FPS, interruption/reconnect, all objects/skins/effects, analytics, and no hitches.

QA Matrix

Matrix	Required cases
Timing	Phase 0; every curated anchor; expected state at 30, 60, and 90 seconds; exact 120-second wrap; multiple loops; 30 FPS; 60 FPS; frame drops.
Gravity independence	Change base gravity; disable ramp; freeze cycle while gravity continues in debug; Purple Fruit for A, B, and both; extreme gravity in day and night.
Synchronization	Two physical devices; mixed frame rates and quality tiers; latency spike; packet loss; background/return; late spectator; replay zero/scrub; bot, ghost, private, and ranked match.
Readability phases	Sunrise, day, sunset, twilight, night, and predawn.
Readability objects	Default, bright, dark, and purple hero; Green Apple; Purple Fruit; Golden Hero Token; hazards; tap effects; character particles; opponent outline; timer; gravity meter; network warning.
Performance	Entry, mid, and high Android tiers; battery saver; thermal throttle; maximum particles; two complex loadouts; night max stars/lights; sunset gradient/clouds; repeated restarts.

Day/Night Analytics

Event	Required properties
Match Environment Initialized	match_id; mode; biome_id; profile_id; starting_phase; environment_seed; graphics_quality; device_performance_class
Match Ended	survival_duration; ending_phase; maximum_gravity; starting_phase; graphics_quality; average_fps; minimum_fps
Environment Resynchronized	drift_amount; correction_method; match_elapsed_time; device_type
Accessibility Setting Used	reduced_motion; reduced_effects; night_visibility; outline_strength

Review whether starting phase correlates with loss/exit rate, low-end devices drop frames in specific phases, sunset changes pickup collection, corrections happen too often, or accessibility settings correlate with outcomes. Correlation triggers investigation; it does not prove cause or justify collecting unnecessary personal data.

Production Acceptance

- Exact seamless 120-second cycle with visibly moving sun and moon.
- Gravity, Purple Fruit, and environment progression are fully independent.
- Two devices maintain the same phase without per-frame server environment updates.
- Night never obscures players, pickups, hazards, or interface; Golden Hero Tokens survive sunset, Green Apples survive green biomes, and Purple Fruit survives twilight/night.
- Saved match data reproduces the environment for replay and spectating.
- Low settings preserve identity; target Android hardware meets frame, thermal, battery, and load-hitch requirements.
- Ambient layers never mask gameplay cues; music never restarts at cycle phases.
- No purchase changes competitive visibility or environment advantage.
- The editor supports a neutral-lighting workflow and a gameplay-lighting preview.
- Gravity is communicated through physical feedback rather than darkness.

- Background reactions remain cosmetic, deterministic where required, and non-distracting.

Future Cosmetic Expansion

Architecture may later support seasonal sun/moon variants, moon phases, aurora, meteor showers, comets, holiday skies, biome celestial bodies, distant storms, eclipses, private-match atmosphere, tournament skies, ranked-season themes, ending-phase victory screens, full-cycle statistics/achievements, secret long-survival cosmetics, spectator presentation, and cinematic replay controls. Every addition remains synchronized, readable, cosmetic, and non-pay-to-win.

PART 3

INFINITY TAPS AND THE LIVING WORLD

The no-loss exploration and fidget game: movement, reactions, zones, discovery, heroes, journal progress, and earned-only status.

15. Infinity Taps: Core Experience

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

Infinity Taps is a no-loss, noncompetitive, local-first exploration and fidget experience where tapping moves the hero and causes the world to react.

CONFIRMED There is no traditional lose condition. Environmental hazards may create spectacle or movement changes, but they must not fail or punish the player.

1 Core Loop

1. The hero begins on the ground in a welcoming environment.
2. The player taps to move left, right, upward, or through the environment based on tap position and current movement state.
3. The world reacts: water splashes, plants move, animals respond, objects produce sound, and heroes acknowledge interaction.
4. The player travels through zones, discovers biomes, encounters heroes and variants, triggers secrets, and fills a journal.
5. Lifetime taps and distance accumulate even when no major discovery occurs.
6. Exclusive Infinity rewards unlock through play and cannot be purchased.

2 Fidget-First Design

The experience must remain satisfying when the player is not pursuing an objective. The tap sound, movement arc, particle response, and environmental acknowledgement should be enough to support repetitive use. Goals enhance the loop but must not turn it into disguised ranked pressure.

3 Currency Rule

CONFIRMED Hero Coins do not spawn as normal pickups in Infinity Taps. The player earns one Hero Coin per 1,000 valid taps.

This rate gives the mode economic acknowledgement without making it the fastest currency farm. The counter should clearly distinguish lifetime taps, current-session taps, and the next Hero Coin threshold.

4 Pause and Persistence

- The player may stop at any moment without losing progress.
- The world can resume near the last safe location or biome checkpoint.
- Temporary boosts may expire during active play but should not punish app interruption.
- Rare encounter state should be saved long enough to avoid frustration from accidental app suspension.
- Offline progress should not generate taps; Infinity Taps is active, not idle.

5 MVP Identity

WORKING The first Infinity Taps prototype should be a small zen playground: one biome, no failure, basic movement, reactive water, bush and tree behavior, one or two animals, a persistent tap counter, several exclusive achievements, and one hidden cosmetic or hero variant.

16. Infinity Taps World Structure, Zones, and Biomes

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Zone Rhythm

CONFIRMED Zones should be approximately two minutes long so a player can experience most of a song or musical arrangement before transitioning.

Approximate time	Purpose	Content
0:00-0:30	Establish the biome	Visual reveal, primary ambient layer, common animals, basic interactions
0:30-1:30	Exploration body	Collectibles, heroes, secrets, environmental chains, temporary boosts
1:30-2:00	Transition	Landmark, intensified layer, departure animation, next-biome foreshadowing

Two minutes is a target cadence, not a rigid timer. A player should be able to linger for an encounter or secret without feeling forcibly removed. The system can measure distance, interaction completion, and time to determine a natural transition.

2 World Progression

Region family	Biomes / layers discussed	Distinctive interaction opportunities
Ground and pastoral	Basic Field, Meadow, Farmland	Grass, flowers, rabbits, fences, crops, puddles
Woodland	Forest, Jungle	Trees, leaves, vines, birds, squirrels, hidden heroes
Elevation	Mountain, cliffs, cloud approach	Wind, echoes, loose rocks, altitude transition
Dry regions	Desert, dunes	Sand bursts, heat shimmer, buried objects, desert animals
Volcanic	Volcanic Region	Lava, ash, obsidian, golem events, heat-reactive effects
Frozen	Arctic, Antarctic, snow fields	Snow piles, ice cracks, snow creatures, aurora ambience
Ocean surface	Beach, Surface Water, Sunlight Zone	Splashes, waves, fish, shells, floating objects
Deep ocean	Twilight, Midnight, Abyssal, Trench / Mariana-inspired depths	Bioluminescence, pressure ambience, rare creatures, hidden ruins
Sky	Cloud Layer, Upper Atmosphere	Cloud puffs, birds, storms, wind, aerial heroes
Space	Orbit, Moon, planets, distant worlds	Low gravity, stars, alien encounters, cosmic tap effects

3 Biome Transition Rules

- Music should crossfade or layer rather than abruptly restart.
- Colors and ambient particles should preview the next region.
- Animals and objects can migrate between compatible border zones.
- The hero's movement state should remain continuous unless a transition is intentionally theatrical.
- The player should understand that they traveled somewhere, not selected a background from a menu.

4 Temporary Movement Boosts

CONFIRMED Discussed Infinity boosts include velocity increases, tap-and-hold "jet engine," anti-gravity, and extra-bouncy movement.

Boost	Behavior	Typical duration	Design guardrail
Velocity Increase	Raises travel speed and momentum.	10-25 seconds	Must remain controllable and not skip meaningful encounters unintentionally.
Tap-and-Hold / Jet Engine	Holding produces sustained thrust and a dramatic engine effect.	10-25 seconds	Clearly separate from standard tap input and prevent accidental activation.
Anti-Gravity	Reverses or changes the normal tap/gravity relationship; taps may force the hero downward.	10-25 seconds	No failure; use as playful traversal, not punishment.
Extra Bouncy	Greatly increases jump height and elastic response.	10-25 seconds	Camera and collision must handle exaggerated movement safely.

5 Rare Encounters

- Alien or unusual creature encounters in space and distant worlds.
- Hero variants tied to specific biomes or weather.
- Extremely rare tap-effect discoveries, including historical 0.001% concepts.
- Environmental storytelling such as footprints, abandoned objects, or creatures leading the player to a hidden area.
- Seasonal worlds that temporarily alter animals, music, and interaction outcomes.

17. Living World Interaction System

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

The central Infinity Taps question is: "I wonder what happens if I tap that." The world should already be moving before the tap; the player interrupts, influences, or plays with a living scene rather than activating a static backdrop.

1 Interaction Depth Model

Layer	Player behavior	World response
Immediate	First tap	Fast readable movement, sound, splash, shake, glance, chirp, or particle response
Escalated	Several taps	Larger animation, relocation, dropped object, altered sound, or social reaction
Persistent	Repeated rapid taps	Special behavior, escape, hiding, mini-event, cooldown, or secret
Contextual	Tap under special conditions	Different result based on biome, weather, time, hero, nearby object, or equipped effect
Chain	Tap creates another object's reaction	Connected sequence that makes the environment feel systemic

2 Animal Behavior

Animal	First response	Repeated / rare response
Rabbit	Looks toward tap, twitches ears, small hop	Hops away, enters burrow, kicks dirt, leads to a hidden object, or returns with an accessory
Bird	Chirps, turns, hops to another branch	Flies away, drops a feather, triggers a flock, lands on the hero, or leads to a nest
Squirrel	Freezes and watches	Climbs, throws an acorn, hides an item, steals a harmless object, or begins a short chase
Frog	Croaks or shifts	Jumps farther, dives into water, lands on a lily pad, or joins a musical sequence
Fish	Changes direction under a ripple	Jumps, scatters a school, reveals a larger fish, treasure, aquatic hero, or rare creature

CONFIRMED Animal interactions must remain playful. Tapping should not visually injure animals.

3 Environmental Objects

Object	Basic reaction	Escalated possibilities
Water	Small splash and ripple	Overlapping waves, larger splash, fish reaction, freezing crack, puddle effect, submerged reveal
Bush	Wiggles	Leaves, berries, insects, or small animals emerge; hidden passage or collectible may appear
Tree leaves	Leaves fall from tapped area	More leaves, fruit, petals, snow, birds, squirrels, or branch motion
Tree trunk / branch	Shake or bend	Object falls, animal relocates, chain reaction begins
Grass / flowers	Bend or sway	Petals, dandelion seeds, bloom sequence, butterflies or bees
Rock / ground	Sound, pebble roll, dust, compression	Fossil, insect, buried object, crack, material-specific secret
Snow	Compresses or puffs	Piles form, melt later, or support a snow creature event
Sand / mud	Burst, dent, or splatter	Tracks, buried item, creature emergence, or temporary pattern

4 Interaction Chains

- A tap shakes a tree; an acorn falls; a squirrel collects it; following the squirrel reveals a hero.
- A water splash startles a fish; the fish jumps; a frog moves; the lily pad exposes a collectible.
- Repeated snow taps create a pile; the pile qualifies for a rare evil snowman emergence.
- Lava pools cool into obsidian; enough qualified interaction can awaken an obsidian golem.
- Flowers release seeds; wind carries them; new flowers bloom elsewhere in the zone.

5 Repeated-Tap Personality

Characters should not loop one reaction forever. A resting hero might open one eye, look around, sit up, become annoyed, throw a pillow, and relocate. An animal might progress from curiosity to playfulness to escape. After the final response, a short cooldown preserves personality and reduces animation spam.

6 Discovery Journal

The journal records biomes, heroes, variants, animals, behaviors, landmarks, secrets, environmental reactions, rare events, and unresolved mysteries. Entries can use partial completion such as "Rabbit Behaviors: 3 of 7" so the player understands that experimentation remains possible without revealing the exact solution.

- World Traveler
- Into the Clouds
- Beyond the Sky
- One Small Tap
- Going Deep
- At the Bottom
- First Contact

18. Heroes, Variants, Encounters, and Collection

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Hero Role

Heroes are playable identity, collectible content, environmental actors, and the bridge between the two products. A hero should be recognizable in competitive play and still feel alive when encountered naturally in Infinity Taps.

2 Encounter Presentation

CONFIRMED When the player finds an unowned hero or variant in Infinity Taps, tapping it should provide an opportunity to purchase or unlock that exact hero or variant.

1. The hero first reacts inside the world.
2. A compact identification card appears without immediately freezing exploration.
3. The card shows hero name, variant name, rarity, ownership state, and acquisition method.
4. A second deliberate action opens the full purchase or unlock view.
5. Closing the view returns the player to the same encounter state where practical.

3 Owned Hero Encounter

- The hero waves, emotes, or acknowledges the player.
- It may briefly accompany the player or trigger a small interaction.
- Repeat encounters can reveal lore, friendship progress, poses, dialogue, or journal details.
- A minor discovery bonus may be granted without turning repeat spawns into currency farming.

4 Owned Base, Unowned Variant

The game must identify the exact variant rather than treating ownership of the base hero as full ownership. Some variants are purchasable after discovery; others require an Infinity challenge, friendship threshold, seasonal event, secret interaction, or direct earn-only condition.

5 Acquisition Categories

Category	Rule	Example use
Standard purchase	Discover and buy with Hero Coins.	Common or catalog heroes
Encounter-exclusive purchase	Can only be bought while or after found in the world.	Biome-linked variants
Challenge unlock	Complete a defined exploration or tap objective.	Traversal or interaction specialist
Friendship unlock	Meet and interact repeatedly.	Personality-driven hero
Secret unlock	Trigger an unusual environmental chain.	Hidden variant
Seasonal encounter	Available during a limited season or event.	Holiday or seasonal form
Infinity-exclusive earn	Cannot be bought; must be earned in Infinity Taps.	Prestige exploration reward

6 Rarity and Discovery

Rarity should describe encounter frequency, acquisition difficulty, or presentation complexity-not competitive power. A rare hero can have more elaborate animation or a demanding discovery condition without superior ranked statistics.

7 Character Identity Systems

- Base hero and named variants.
- Idle animation, movement style, tap response, victory pose, and entrance.
- Short lore card plus discoverable notes.
- Compatible particle, tap, trail, and profile presentation.
- Optional friendship or encounter history in Infinity Taps.

19. Velvet Hex Character, Variant, and Cast Production Reference

Source basis: *Game Design Bible (1)* / duplicate *Game Design Bible* pp. 14-17; *TAM Design Bible* pp. 18-19; current workspace skin catalog and supplied art assets.

CHARACTER PURPOSE Characters are the emotional and collectible face of TAM. They must be instantly recognizable in motion, readable on a phone, expressive under squash-and-stretch animation, and mechanically equivalent in competitive play.

Velvet Hex Canon

Velvet Hex - A stitched tuxedo fiend that trades cuddles for tiny curses.

Facet	Production requirement
Core silhouette	Tuxedo-cat body; plush/slime-like triangular mass; cat ears; two demon horns; pointed/spade-like tail; button/X eyes; white tuxedo chest; purple bow tie; small fangs where readable.
Personality	Cute, mischievous, strange, cuddly, skeptical, faintly dangerous, and expressive without cruelty.
Movement identity	Squash, stretch, wobble, float, and splat-like landings while the silhouette remains identifiable at native sprite scale.
Material flexibility	The body may read as stitched plush, slime, ice, toxic ooze, royal fabric, molten matter, spectral material, arcade energy, or costume without losing the anchor features.
Competitive footprint	Every skin and hero uses the approved equivalent collision/gameplay footprint unless a future non-monetized ability system is separately balanced and declared.
Pixel-art standard	Retro 1990s arcade-inspired pixel art, small native sprites, visible clusters, limited theme palette, crisp nearest-neighbor integer scaling, true alpha, and no anti-aliased/painterly filter treatment.

Standard Six-Frame Jump Sheet

Frame	Pose	Purpose
1	Neutral / ready	Baseline idle or airborne-ready silhouette.
2	Compression / squash	Anticipation immediately before upward force.
3	Launch / stretch	Elongation as the impulse begins.
4	Rising / airborne	Body regains its recognizable shape during ascent.
5	Apex / suspended	Readable hang-time pose before descent.
6	Landing / impact	Squash plus theme-specific particles or impact treatment.

PRODUCTION LOCK All variants use identical frame dimensions, baseline, pivot points, frame order, collision reference, and export layout so a texture/skin swap does not require animation-code changes. Final native dimensions, proportions, and per-frame pixel budget remain an art-production decision until formally approved.

Named Velvet Hex Variant Register

Variant	Defining treatment	Evidence / status
Classic Velvet Hex	Black-and-white tuxedo slime/plush demon; purple horns, button/X eyes, bow tie, white chest, fangs, pointed tail.	Canonical silhouette anchor; current default catalog item.
Scarlet / Vampire Hex	Dracula-inspired red/black/purple coat or trench treatment; red horns; vampire teeth; tuxedo chest and tie retained.	Documented production variant; current store ID not present.
Beach Hex	Bright tropical/beachwear treatment while retaining chest, bow tie, silhouette, and animation layout.	Documented production variant; current store ID not present.
Toxic Hex	Neon toxic-green ooze, bubbles, drips, green ear highlights, contaminated horns, toxic landing splash.	Documented and current 100-Hero-Coin catalog item.
Golden / Royal Hex	Gold and purple royal trim, crown/horn ornament, cloak accents, and gold-edged silhouette.	Documented; current Golden Hex catalog item at 100 Hero Coins.
Arcade Hex	Neon cyan/magenta/purple, arcade icons, scanline/energy motifs, and pixel sparks.	Documented and current 100-Hero-Coin catalog item.
Frozen Hex	Deep ice coverage, shard horns, ice-cube eyes, frosted edge buildup, blue bow tie, snow/ice particles, readable tuxedo chest.	Documented and current 100-Hero-Coin catalog item; production correction applies.
Molten Hex	Glowing cracks and molten/fire material while preserving silhouette/readability.	Candidate promoted into current 100-Hero-Coin catalog.
Ghost Hex	Pale spectral/translucent treatment without reducing trackability or collision readability.	Candidate promoted into current 100-Hero-Coin catalog.
Corrupted Hex	Controlled pixel corruption/glitch accents localized around a readable body core.	Candidate promoted into current 100-Hero-Coin catalog.
Strawberry Slime	Pink/red seeded slime identity with playful food-material treatment.	Candidate promoted into current 100-Hero-Coin catalog.
Abyssal Hex	Deep-sea teal bioluminescence and self-lit detail.	Current 100-Hero-Coin catalog item; earlier Infinity-exclusive Abyssal Velvet concept creates an acquisition conflict to resolve.
Meadow Hex	Soft green field/floral treatment.	Current 100-Hero-Coin catalog item; earlier Infinity-exclusive Meadow Velvet concept creates an acquisition conflict to resolve.
First Contact Hex	Alien/contact treatment with bright green accents.	Current 100-Hero-Coin catalog item; earlier Infinity-exclusive First Contact concept creates an acquisition conflict to resolve.

Variant	Defining treatment	Evidence / status
Cosmic Hex	Animated/variable nebula material and premium cosmic presentation.	Current 200-Hero-Coin catalog item.
Prismatic Hex	Completionist prestige treatment.	Current earned-only item; price -1; never purchasable or bundled.

OPEN ACQUISITION CONFLICT The current store sells Abyssal, Meadow, and First Contact variants, while the Original Expanded Bible uses closely matching names for Infinity-exclusive rewards. Do not silently treat both as the same item. Resolve by renaming, making distinct designs/IDs, or changing acquisition before exclusivity claims ship.

Frozen Hex Production Correction

Frozen Hex uses the same complete six-frame sheet and clean transparent/light presentation as approved Classic, Scarlet, Beach, Toxic, Royal, and Arcade sheets. Dark atmospheric backgrounds, missing frames, inconsistent pivots, shallow ice coverage, or concept-sheet framing are concept-art errors, not production standards.

Cast and Entity Role Register

Entity family	Role and behavior	Player-facing status
Velvet Hex	Brand anchor, playable hero, collection identity, reaction performer, and variant platform across both products.	Canonical / partially implemented through current skins and assets.
Future named heroes	Distinct silhouettes, personality, lore, reaction trees, exact variants, acquisition source, and equivalent Competitive footprint.	Roster TBD; do not fabricate launch names or promise count.
Evil Snowman	Harmless cosmetic rare-event creature eligible after a qualifying snow accumulation; appears, performs, and departs/melts.	Candidate environmental entity, not a playable hero unless separately promoted.
Obsidian Golem	Harmless cosmetic rare-event creature eligible after a qualifying magma/obsidian pool; poses and crumbles.	Candidate environmental entity, not a playable hero unless separately promoted.
Tiny companions	Slime, ghost, frog, alien, snow creature, fire sprite, bat, cloud, or similar brief followers/reaction performers.	Candidate cosmetic/Infinity content; compatibility and permanence defined per item.
Ambient animals	Rabbit, squirrel, fish, frog, bird, whale, butterfly, deep-sea creature, and biome wildlife with playful staged reactions.	Infinity living-world content; never visually injured by taps.
Rare travelers / aliens / spirits	Named or mysterious encounters that may recur across zones/worlds, reveal lore, or unlock earned content.	Candidate discovery content; exact roster and odds open.
Bots / ghosts	Competitive opponent presentations governed by separate fairness, disclosure, behavior, and rating rules.	Not collectible heroes merely because they use a hero skin.

Every named entity record should include stable entity ID, display name, entity class, role, products, biome/world eligibility, variants, personality tags, animation set, reaction states, audio set, acquisition/discovery policy, rarity, ownership behavior, Journal behavior, accessibility notes, and content version.

20. Infinity Taps Detailed World, Discovery, and Implementation Reference

Source basis: Original Expanded Bible pp. 45-79 and 81; TAM Game Design Bible v1.0 Sections 13-16; Game Bible v1.0 Infinity chapters.

LOCKED Infinity Taps is a no-loss, noncompetitive, local-first tapping, fidget, and exploration product. It is not survival with failure removed. It has no lives, rising survival gravity, health bar, forced timer, PvP pressure, idle punishment, or traditional loss screen.

Emotional Contract

The player opens Infinity to tap, move, regulate attention, watch satisfying reactions, discover a living world, and build permanent identity without being judged. Ordinary taps must feel good even when no achievement, currency, or rare event occurs. Goals enhance curiosity; they must not turn the product into disguised ranked pressure.

- Every valid tap produces readable motion, sound, haptic, particle, character, or environmental acknowledgement.
- The player may stop at any moment and resume without failure, lost progress, or a scolding return flow.
- The world is already alive before the player touches it; tapping interrupts, influences, or plays with an existing scene.
- Discovery rewards observation, repetition, experimentation, patience, direction, and context.
- Some of the shared universe's most interesting cosmetics belong exclusively to noncompetitive exploration.

Core Loop

1. Resume the last safe biome, zone, and local state.
2. Tap relative to the hero to move left, right, upward, downward where appropriate, or through an environment-specific movement state.
3. Increment session and lifetime tap totals after input validation.
4. Move, squash, stretch, wobble, or otherwise respond immediately.
5. Trigger the equipped tap effect and character-particle response.
6. Cause water, plants, terrain, animals, objects, heroes, structures, weather, or hidden triggers to react.
7. Advance distance, zone, biome, altitude, depth, discovery, encounter, journal, achievement, and reward checks.
8. Save safely at intervals and when the app backgrounds or exits.

Movement Languages

Context	Tap-driven movement	Presentation intention
Ground	Hop, bounce, bound, and bias left/right by tap position.	Forgiving and rhythmic; never trap the hero in geometry.
Mountain / vertical terrain	Climb, rebound, launch from ledges, and use short boosts.	Direction and height feel exploratory, not punishing.
Water	Swimming bursts; taps can move horizontally, upward, or deeper.	Bubbles, drag, current, and creature reactions replace standard falling.
Low gravity / space	Long drifting arcs with smaller corrections.	Slow particles, floating accessories, and broad movement paths.
Velocity boost	Short 10-25 second speed state.	A playful remix of movement, audio, camera, and reaction timing.
Jet-engine hold	Temporary tap-and-hold sustained thrust.	A deliberate exception to repeated tapping; biome/boost specific.
Anti-gravity / extra-bouncy	Temporary altered arc and rebound behavior.	No competitive fairness requirement, but always readable and comfortable.

Candidate Infinity Boost Backlog

CANDIDATE Boosts normally last approximately 10-25 seconds and temporarily remix movement, reaction timing, camera feedback, particles, and one music layer without introducing failure. Exact durations and stacking remain open.

Boost	Movement / interaction change	Audio and safety intent
Velocity Increase	Raises horizontal speed and momentum so the hero races through part of a biome.	Faster rhythmic layer; camera remains comfortable and interactions remain reachable.
Jet Engine	Holding the screen creates continuous upward thrust; a fuel/heat presentation may limit the state.	Rising synth; no overheat failure, damage, or lost progress.
Anti-Gravity	Reverses the ordinary impulse interpretation so taps push downward instead of upward.	May reverse or invert part of the motif; controls are explicitly taught.
Extra Bouncy	Greatly increases jump height and ground rebound for absurd launches.	Spring percussion; geometry prevents trapping or camera sickness.
Heavy Gravity	Creates smaller jumps but more dramatic landings and environmental reactions.	Heavier impact layer; never becomes survival pressure.
Tap Burst	Every tap emits a short interaction shockwave affecting nearby leaves, water, bushes, and animals.	Clear radius and cooldown; playful reactions only.
Moon Bounce	Lowers gravity for slow, exaggerated arcs and gentle correction.	Wide, airy music layer; never stalls the player permanently.
Magnetism	Pulls nearby Infinity-specific collectibles or hidden objects toward the hero.	Does not affect competitive pickups, paid rarity, or exclusive eligibility outside its declared content.

Tap Counting and Hero Coins

CONFIRMED Infinity normally awards one Hero Coin for every 1,000 valid active taps. Hero Coins do not spawn as ordinary Infinity pickups. Preserve the remainder across sessions.

- Show lifetime taps, current-session taps, and progress toward the next 1,000 threshold distinctly.

- Count approved touch/mouse events only while the world is active; menus, overlays, duplicate events, and implausible automation do not count.
- Do not hard-limit normal fast fidget tapping; use generous plausibility analysis.
- Offline play increments local progress, but trusted currency reconciliation must prevent save editing from minting unlimited server value.
- No offline idle accumulation exists: Infinity rewards active tapping, not elapsed time while closed.

Zone Rhythm

CONFIRMED A typical zone targets approximately two minutes of ordinary traversal so the player can experience most of its musical arrangement. This is a cadence, never a timer or forced transition.

Working time band	Zone role
0:00-0:30	Establish the biome, movement language, music, wildlife, and obvious environmental interactions.
0:30-1:30	Exploration, collectibles, heroes, secrets, repeated-tap depth, interaction chains, and temporary boosts.
1:30-2:00	Visually and musically preview/transition toward the next zone while preserving movement and avoiding a hard menu cut.

World and Biome Framework

A biome is not a background. It defines movement feel, signature interactions, wildlife, palette, ambience, music instrumentation, secret vocabulary, hero pool, boosts, landmarks, rare encounters, and exclusive rewards.

World layer	Canonical / proposed content
Starting Earth surface	Welcoming Meadow/field with grass, flowers, hills, clouds, butterflies, water, bushes, trees, rabbits, and a first hero encounter.
Surface expansion	Farmland, Forest, Mountain, Desert, Volcanic region, Beach, Crystal Caverns, Neon City, Mushroom Grove, Snow/Arctic, and Antarctic as a distinct polar region.
Ocean	Surface water, sunlight zone, twilight zone, midnight zone, abyssal zone, and deepest trench with vents, rare wildlife, ancient structures, and exclusive rewards.
Sky	Birds, balloons, aircraft, storms, rainbows, floating islands, sky heroes, and rare shooting stars.
Space	Satellites, stations, meteors, comets, asteroids, astronauts, alien ships, signals, moons, and low-gravity travel.
Other worlds	Mars-like desert, ice, slime, crystal, gas-giant platforms, machine world, low-gravity garden, dark void, candy-colored alien world, and future seasonal worlds.

SCOPE The proposed early eight-biome set (Meadow, Forest, Snow, Beach, Crystal Caverns, Neon City, Mushroom Grove, Space) is a content recommendation, not permission to build eight shallow biomes before the one-biome vertical slice is excellent.

Living World Interaction Depth

The central question is: 'I wonder what happens if I tap that.' Reactions support multiple thresholds, cooldowns, context conditions, randomized alternatives, and secret combinations. Not every object needs maximum depth at launch, but the data model must support it.

Depth level	Example behavior
First tap	Immediate acknowledgement: splash, wiggle, blink, sound, leaf fall, look, or small motion.

Depth level	Example behavior
Repeated taps	Escalation: larger response, annoyance, play, movement, transformation, accumulation, or a different animation.
Context	Time, direction, current biome, equipped effect, owned hero, weather, nearby object, previous discovery, or cooldown changes the result.
Chain	One reaction affects another object or creature and produces a discoverable sequence.
Secret / rare	A hidden threshold, unusual order, location, or long-term condition unlocks an encounter, journal entry, or earned item.

- Shake a tree; an acorn falls; a squirrel collects it; following the squirrel reveals a hero.
- Splash water; a fish jumps; a frog moves; its lily pad exposes a collectible.
- Build a snow pile; cross a threshold; perform one qualified Evil Snowman check.
- Accumulate magma; cool it into obsidian; awaken a cosmetic Obsidian Golem; let it crumble harmlessly.
- Release flower seeds; let wind carry them; grow a temporary patch elsewhere in the zone.

Animals and Hero Reactions

LOCKED Animal interactions are playful. Tapping never visually injures an animal, creates failure, or turns curiosity into punishment.

Characters should not replay one reaction forever. A resting hero may open one eye, look around, sit up, become annoyed, throw a pillow, then relocate. An animal may move from curiosity to playfulness to escape. Cooldowns preserve personality and prevent spam.

When an unowned hero or exact variant appears, the first interaction happens in-world. A compact identification card then shows hero, variant, rarity, ownership, and acquisition method. Only a second deliberate action opens purchase or unlock details. Closing should return to the same encounter where practical. Owned heroes may wave, accompany briefly, reveal lore, friendship, poses, or journal details.

Discovery Journal

- Biomes and worlds visited; zones and landmarks reached.
- Animals found and distinct behaviors witnessed, including partial progress such as Rabbit Behaviors: 3 of 7.
- Heroes and exact variants encountered, owned, purchased, earned, or still mysterious.
- Environmental reactions, interaction chains, rare events, secret locations, and unresolved mysteries.
- Highest altitude, greatest depth, lifetime taps, session records, worlds traveled, and special movement states.
- Cosmetics found, acquisition source, permanent ownership, and mode-exclusive status.

Common entries may show clear requirements. Secret entries may use silhouettes, riddles, partial hints, or remain completely hidden until first discovery. The Journal should encourage observation without reducing every mystery to a checklist.

Rare Discovery and Pity

Encounter tiers may include common ambient, uncommon named, rare traveler, biome-exclusive, seasonal, secret, and legendary. Extremely low odds can create wonder, but cannot create plausible years of engagement with no path to success.

OPEN The original expanded design gives an example check every 100 valid taps with a 0.1% base chance (roughly one success per 100,000 taps before pity). The exact check interval, base odds, pity curve, drought reset, and category-specific protection remain open and must be simulated before launch.

- Roll on meaningful intervals or threshold crossings, not every rendered particle.
- Track eligibility, time/taps since last success, and hidden pity progress.
- Increase opportunity after a long qualifying drought without displaying exact secret thresholds.
- Use idempotent grants so reconnects and retries cannot duplicate rare ownership.
- Store discovery source and condition so the collection can explain how an item was earned.

HISTORICAL TEST CANDIDATES The Original Expanded Bible also proposes 0.001% per eligible tap, slowly increasing chance after 50,000 eligible taps, and a guaranteed configured maximum. Another framing rolls once per 100 valid taps. These examples are not mutually approved and must be simulated as competing models, not combined accidentally.

A sustained rate around 15-20 taps per second is an example integrity-review threshold, not an automatic punishment threshold. Fast fidget players must not be treated as cheaters merely for tapping rapidly; use device/input evidence, duration, variability, menu state, server plausibility, and repeated patterns. Offline Hero Coin and premium/exclusive unlock evidence may remain pending until trusted synchronization completes.

Infinity-Exclusive Rewards

LOCKED Designated Infinity skins, variants, trails, tap effects, particles, frames, badges, titles, and companions cannot be purchased with money, Hero Coins, a pass, a supporter bundle, or an advertisement.

Unlock family	Examples
Tap milestones	Lifetime totals, session totals, long fidget sessions, unusual rhythms.
World discovery	Every starting-field landmark; every Earth biome; all ocean depths; all launch planets.
Encounter	Hidden forest spirit, named creature, rare hero variant, unusual weather, cosmic visitor.
Depth / altitude	Deepest trench, volcanic chamber, cloud line, orbit, moon, planet.
Collection	Every creature/behavior in a biome; complete chain set; Journal completion.
Conditional secret	Tap a meteor in a specific state; follow an animal chain; combine effects or environmental conditions.

Items display source text such as 'Abyssal Velvet - Reached the deepest trench in Infinity Taps - Cannot be purchased.' They may be wearable in Competitive TAM after being earned, but the competitive shop never sells them.

Proposed Infinity-Exclusive Seed Catalog

Reward	Type	Proposed unlock requirement
Meadow Velvet	Skin	Discover every landmark in the starting field.
Forest Spirit	Trail	Encounter the hidden forest spirit.
Volcanic	Tap effect	Reach the deepest volcanic chamber.
Abyssal Velvet	Skin	Reach the deepest ocean trench.
Bioluminescent	Trail	Discover five deep-sea creatures.
Moon Dust	Tap effect	Land on the Moon.
Alien Signal	Effect	Witness a rare alien transmission.
First Contact	Skin	Encounter an alien character.
World Traveler	Profile frame	Visit every launch-version biome.
Endless Ripple	Effect	Reach 100,000 Infinity Taps.
Million Tap	Aura	Reach 1,000,000 Infinity Taps.
Cosmic Velvet	Skin	Discover all initial space regions.

PROPOSED / CONFLICT CHECK These are seed concepts, not current catalog commitments. Meadow, Abyssal, and First Contact names conflict with currently purchasable store variants; exclusivity requires distinct IDs/designs or a corrected store/acquisition decision before release.

On rare-world spawn, save the opportunity immediately, leave enough time to collect it, preserve it across accidental close where feasible, grant permanently, suppress meaningless duplicates, and attach a unique transaction key such as `infinity\_unlock:skin\_velvet\_abyssal`. The store service rejects any price assignment where `purchasable == false`.

Save and Shared-Account Boundary

- Current world, biome, zone/checkpoint, horizontal distance, altitude, depth, highest altitude, and greatest depth.
- Lifetime/session taps and remainder toward the next 1,000-tap coin.
- Discovered biomes, worlds, heroes, variants, animals, behaviors, landmarks, events, secrets, and cosmetics.
- Rare-discovery eligibility, pity, last qualifying check, last successful discovery, and pending grants.
- Product source, ruleset/content version, local timestamp, cloud revision, and idempotency IDs.

The shared user owns an eligible hero once, but Competitive rating and season state remain Competitive-only; Infinity world and Journal progress remain Infinity-only. Cross-device merge must not blindly choose the larger or newer save when that would duplicate currency or erase discoveries.

Infinity Vertical-Slice Release Gate

- No input, hazard, geometry, boost, or environmental event produces a loss screen.
- Directional movement is satisfying with no reward layer enabled.
- Water splashes, a bush wiggles, tree leaves fall, and repeated interactions vary.
- At least one animal has three distinct interaction stages.
- One found hero or variant uses the two-step in-world identification/acquisition flow.
- Lifetime taps persist and one Hero Coin is awarded exactly once at 1,000 valid taps.

PART 4

PROGRESSION, IDENTITY, AND ECONOMY

The shared collection ecosystem: currencies, achievements, badges, challenges, cosmetics, tap effects, particles, and ethical monetization.

21. Progression, Currencies, and Reward Economy

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Progression Dimensions

Dimension	Competitive TAM	Infinity Taps	Shared profile
Skill	Glicko rating, tier, wins, survival, records	Not ranked	Best milestones
Activity	Matches, valid taps, challenges	Lifetime taps, distance, sessions	Total taps with source detail
Discovery	Pickups, event modes, achievements	Biomes, animals, heroes, secrets	Journal and collection summary
Collection	Cosmetics, heroes, badges	Exclusive finds, heroes, variants	Owned catalog
Season	Ranked climb and season rewards	Seasonal worlds or encounters	Season history

2 Hero Coin

Hero Coin is the main cosmetic currency. It can be earned through achievements, competitive rewards, controlled pickups, challenges, seasonal progression, and Infinity's one-per-1,000-valid-taps rule. The currency must not purchase ranked power.

3 Economy Sources

- Achievement rewards.
- Daily and weekly challenge rewards.
- Match completion and limited performance bonuses.
- Seasonal ladder milestones.
- Golden Hero token pickups under capped rules.
- Infinity tap milestones.
- Starter purchase and other clearly priced cosmetic bundles.

4 Economy Sinks

- Heroes and purchasable variants.
- Tap effects and color variants.
- Character particle effects and trails.
- Profile frames, username treatments, badges where not achievement-exclusive, and emotes.
- Entrances, victory poses, landing effects, arena themes, and presentation items.

5 Economy Guardrails

- No direct purchase of rating, survival force, tap validity, movement speed, damage, life, regeneration, pickup priority, or matchmaking advantage.
- Do not make basic visual readability a paid feature.
- Infinity-exclusive earnables remain unavailable for purchase.
- Seasonal exclusivity should be disclosed before purchase and not falsely returned as identical content later.
- Currency pricing must be understandable; avoid manipulative multi-currency conversion chains.

6 Long-Term Collection

The collection should show how an item was obtained: shop, achievement, season, ranked reward, Infinity discovery, secret, or supporter bundle. Source labeling strengthens prestige and helps players understand what remains possible.

22. Achievements, Badges, and Challenges

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

CONFIRMED Achievements are permanent long-term goals. Daily and weekly challenges rotate. Badges are permanent profile collectibles.

1 Profile Display Rules

- One primary badge appears beside the username and in prominent competitive presentation.
- Up to three additional showcase badges appear on the full profile.
- Tiered achievement families display the highest earned tier by default.
- Earlier tiers remain in collection history.
- Server validation must ensure a loadout can display only badges the user actually earned.

2 Achievement Categories

Category	Purpose	Examples discussed
Fruit	Reward pickup behavior, risk, and unusual poison outcomes.	Life of Poisonous Fruit, Fruit Salad, Toxic Appetite, Poison-Proof, Apple a Day
Survival	Reward duration, control, recovery, and repeated runs.	First Flight, Learning to Fly, Gravity Defier, Sky Eternal, Close Call
Coins and Shop	Reward earning, spending, and cosmetic collection.	Coin Collector, Golden Hoard, Token Hunter, Big Spender, Shop Regular, Fully Dressed
Multiplayer	Reward first wins, sustained competition, and opponent variety.	First Victory and additional catalog entries
Progression	Reward onboarding and long-term milestones.	Getting Started and tiered milestone families
Secret	Reward experimentation and unexpected behavior.	Was That Worth It?, Floor Inspector, Maximum Poison, Perfectly Balanced, Velvet's Nine Lives, Too Greedy
Infinity Exploration	Reward travel, depth, space, behaviors, and discoveries.	World Traveler, Into the Clouds, Beyond the Sky, One Small Tap, Going Deep, At the Bottom, First Contact
Seasonal	Provide time-bound goals without removing permanent progress.	Season participation, rank, world-event, and collection achievements

3 Current Implementation Note

CONFIRMED The deployed backend health response reports 55 achievement definitions.

The design discussion preserved many exact names but not every threshold and reward. Engineering should export the authoritative achievement catalog from shared/achievements.js and reconcile it against this bible before content lock. The count of 55 should not be treated as a permanent ceiling.

4 Tiering

Achievement families may use Bronze, Silver, Gold, and Mythic or equivalent tiers. Thresholds should reflect real behavior distributions. A tier should never require a paid item unless a free acquisition path exists and the achievement is explicitly about collection rather than competitive skill.

5 Challenge Structure

- Daily challenges should be short, varied, and completable without extreme luck.
- Weekly challenges may span multiple modes or encourage deeper goals.
- Infinity challenges should avoid forcing competitive play; competitive challenges should not require Infinity-exclusive content.
- Challenges may award Hero Coins, progress, profile items, or limited cosmetics.
- Secret achievements should not appear as explicit checklist chores before discovery.

6 Reward Philosophy

Achievements should reward behavior, mastery, experimentation, and identity. The strongest rewards are not merely currency; they are badges, effects, skins, encounters, profile treatments, and stories the player can associate with a specific accomplishment.

23. Cosmetics, Tap Effects, and Character Particles

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Cosmetic Categories

- Hero skins and variants.
- Character particle effects and auras.
- Tap effects.
- Movement trails, water trails, and space trails.
- Spawn, entrance, landing, elimination, and victory animations.
- Arena and background themes.
- Fruit skins and pickup presentation.
- Profile frames, badges, username colors, titles, and emotes.
- Ambient sound themes and idle animations.
- Companion creatures or pets where they do not clutter competitive readability.

2 Established Tap Effects

Effect	Behavior	Price / unlock note
Pulsing Rings	Expanding rings at the tap point. Basic purple; custom color picker; RGB-changing variant.	Purple: 100 Hero Coins. Additional custom colors: 50. RGB: 300.
Rainstorm	Rain falls around the tap. Extremely rare lightning can strike and temporarily scorch the ground.	200 Hero Coins. Historical lightning chance: 0.001%.
Snowfall	Snow falls and piles temporarily before melting. Qualified piles can produce an evil snowman.	200 Hero Coins. Rare event after pile threshold.
Magma / Lava	Lava or magma pools form and cool; qualified buildup may produce an obsidian golem.	Working price: 200 Hero Coins.
Fire	Flame response at the tap point with possible color or theme variants.	Base concept: 75 Hero Coins; variants may include purple, blue, ghost, pixel, or golden fire.
Autumn Leaves	Leaves fall and collect; a sufficient pile can burst or trigger a larger response.	Working price: 200 Hero Coins.

3 Suggested Tap Effect Catalog

These are idea-bank candidates rather than launch requirements.

- Bubble Pop; Pixel Lightning; Starfall; Galaxy Portal; Ghost Hands; Slime Splash; Ink Blot; Paint Burst.
- Flower Bloom; Mushroom Growth; Crystal Growth; Mini Earthquake; Water Splash; Ocean Wave; Sand Burst.
- Candy Explosion; Confetti Cannon; Musical Notes; Retro Arcade Hit; Glitch Effect; Shadow Creatures.
- Tiny Footprints; Magic Runes; Black Hole; Cloud Puff.

4 Character Particle Concepts

- Purple smoke.
- Golden sparkles.
- Slime droplets.
- Pixel stars.
- Poison bubbles.

- Tiny ghosts.
- Lightning pixels.
- Falling flower petals.

5 Effect Design Rules

- Tap effects respond to the tap location; character particles remain attached to the hero.
- Rare events should be delightful but not required for basic progression.
- Competitive effects must not obscure the ground, opponent, pickups, or collision boundaries.
- Persistent effects should have strict object limits and cleanup rules.
- Color customization must maintain contrast and avoid flashing hazards.
- Effect combinations should be previewable before purchase or equip.

6 Infinity Exclusives

CONFIRMED Effects, skins, trails, variants, badges, frames, colors, companions, emotes, idle animations, and other rewards designated as Infinity-earned cannot be purchased.

Examples discussed include Meadow Velvet Skin, Forest Spirit Trail, volcanic-themed rewards, water trails, space trails, and hidden variants. Exact final names should be locked during content production.

24. Expanded Tap Effects and Character Particle Catalog

Source basis: Original Expanded Bible pp. 5-44; TAM Game Design Bible cosmetic catalog; current store catalog for shipped effects.

Cosmetic System Boundary

Tap effects appear at the valid tap location. Character particles attach to and follow the hero. Trails, idle components, jump/fall/landing bursts, environment variants, and elimination presentation may be packaged within one character-particle slot. Every component is cosmetic.

- Never alter jump force, gravity, movement speed, collision, pickup range, spawn probabilities, score, rating, coin earnings, or rare-discovery odds.
- The local player may see full intensity; other players receive a reduced, readable version or may disable remote cosmetics.
- Provide density, opacity, flash, screen-shake, sound, haptic, and other-player effect controls.
- Pool repeated objects, cap particles and accumulation zones, shorten lifetimes, merge nearby zones, and degrade gracefully on low-tier devices.
- A rare creature or environmental event creates delight, never power, currency, damage, path blocking, or a hidden competitive signal.

Established Environmental Tap Effects

Effect	Suggested price	Detailed behavior
Pulsing Rings	100 base; 50 per custom color; 300 RGB	Expanding fade ring. Custom hue/saturation/brightness/RGB/darkness/opacity. RGB modes: Random, Smooth Rainbow, Custom Palette. Avoid flashing.
Rain	200	Localized shower. Rare 0.001% weather-lightning concept creates controlled flash, thunder, temporary scorch, smoke, and reactions without damage.
Snow	200	Temporary dusting/pile stages. Repeated taps accumulate; capped pile melts. One qualifying threshold check may produce a harmless Evil Snowman.
Magma	200	Splash, puddle, pool, cooling, and obsidian stages. One qualifying threshold check may produce an Obsidian Golem that poses then crumbles.
Fire	75	Affordable quick flame, sparks, embers, and smoke; repeated taps may grow a capped temporary flame.
Autumn Leaves	200	Color/shape/rotation variety, temporary piles, and a landing/movement burst that never changes real movement.

Additional Tap Effect Backlog

The following catalog is candidate content, not a launch commitment. Approximately one-third of effects should have accumulation or rare-event depth; if every effect summons something, discovery loses meaning.

Effect	Coins	Immediate response	Deeper behavior
Bubble Pop	100	Translucent bubbles float and can be tapped again to pop.	Rare giant bubble may briefly carry the hero cosmetically.
Pixel Lightning	175	Arcade electrical bolt chains to nearby surfaces; repeated taps build static.	Maximum charge creates a chain and may reveal a tiny electric creature; visually distinct from weather lightning.
Starfall	200	A falling star strikes the tap point and leaves a temporary glow.	Consecutive taps form constellations; completed patterns may trigger a cosmetic event.

Effect	Coins	Immediate response	Deeper behavior
Galaxy Portal	250	A small cosmic portal opens, spirals particles inward, and closes.	Rare alien hand, eye, or creature may peer through.
Ghost Hands	175	Translucent hands or paws wave, clap, point, or grab harmlessly.	Nearby taps can summon a temporary ghost audience.
Slime Splash	150	Colored slime splashes and temporarily sticks to surfaces.	Repeated taps grow a puddle; landing creates only a visual bounce; rare tiny slime may emerge.
Ink Blot	125	Random ink shapes spread, drip, and fade.	Rare forms resemble faces, creatures, or symbols.
Paint Burst	100 / 200 / 300	Single-color, custom-palette, or rainbow paint temporarily coats the environment.	Multiple taps paint a larger temporary area; color always fades.
Flower Bloom	200	Flowers grow rapidly at the tap point and build a temporary garden.	Butterflies or a rare giant strange flower may appear.
Mushroom Growth	200	Mushrooms vary in size, color, and shape; clusters release spores.	Rare mushroom creature wakes and walks away.
Crystal Growth	225	Crystals rise, grow with repeated taps, shimmer, then dissolve.	Rare crystal guardian or tiny crystal animal may appear.
Mini Earthquake	175	A cosmetic crack, dust, and stones expand with repeated taps.	Harmless rock creature may peek out; screen shake is optional and off by default.
Water Splash	150	Droplets splash and form temporary puddles with reflections.	Landing splashes; rare frog or water creature may appear.
Ocean Wave	225	A wave and foam sweep outward; consecutive taps make it slightly larger.	Tiny boat, fish, or message bottle may ride a rare wave.
Sand Burst	175	Sand erupts and forms temporary piles the hero can visually scatter.	Scarab, buried object, or miniature sand creature may emerge.
Candy Explosion	200	Wrapped candy, gumdrops, jellybeans, and lollipops bounce and fade.	Rare candy monster or pinata-like creature may appear.
Confetti Cannon	100	Themeable confetti creates a performance-friendly celebration.	Sustained tapping builds a larger but capped burst.
Musical Notes	150 per sound set	Tap position changes pitch; notes pair with piano, chiptune, bells, drums, or slime-pop sets.	Fast tapping can create simple melodies; strong fit for fidget play.
Retro Arcade Hit	125	Localized POW/TAP text, pixel stars, score-like numbers, CRT touches, and comic bursts.	Premium variant may randomize retro words and symbols; numbers never represent real score.
Glitch Effect	250	Localized pixel fragmentation and RGB separation distort the tapped area.	Repeated taps create temporary corruption; rare glitch character may appear; never distort the whole screen.
Shadow Creatures	225	Small shadows crawl, wave, run, or hide.	Several taps form a group; rare one becomes a named encounter.
Tiny Footprints	75 each / 250 bundle	Cat, demon, slime, bird, dinosaur, or alien prints form a temporary trail.	Prints remain cosmetic and fade.
Magic Runes	225	Runes rotate and fade; nearby runes can connect into a magic circle.	Completed circle may summon a harmless creature or portal.
Black Hole	300	A small void pulls cosmetic particles inward and closes quickly.	Repeated taps strengthen visual pull; strange object may be expelled; never affect gameplay objects.
Cloud Puff	125	Small clouds drift and can form larger temporary groups.	Rare face, animal shape, or tiny storm cloud may form.

Effect	Coins	Immediate response	Deeper behavior
Fireworks	200 / 250 / 350	Delayed fireworks use basic, custom-color, or shape-bundle variants.	Repeated taps coordinate a display; avoid flashes that compromise comfort.
Miniature Meteor	225	A small meteor creates a temporary crater and smoke.	Repeated taps create a shower; rare alien object or creature may emerge.
Hearts and Affection	75	Pixel, gothic, broken, slime, or rainbow hearts float upward.	Nearby characters may react positively.
Bat Swarm	175	Tiny bats scatter from the tap point and build a larger capped swarm.	Rare named bat companion may appear briefly.
Tiny Companion Reaction	250-400	A tiny slime, ghost, frog, alien, snow creature, or fire sprite performs one quick animation.	Premium complexity depends on character animation and context support.

Price band	Content expectation
75-100	Simple, immediate, affordable effect.
125-175	Additional movement, variation, theme, or modest interaction.
200-250	Accumulation, environment memory, chain, or rare-event eligibility.
300-400	Highly customizable, complete, or elaborate premium expression.

First-priority candidates because they are distinct without excessive production cost: Bubble Pop, Slime Splash, Musical Notes, Flower Bloom, Retro Arcade Hit, Magic Runes, Galaxy Portal, Tiny Footprints, Confetti Cannon, and Glitch Effect.

Character Particle Seed Catalog

Effect	Coins	Components / environment behavior
Purple Slime Droplets	100	Stretching droplets; ground splatter; underwater slime bubbles.
Shadow Smoke	150	Dark curling trail with purple highlights and idle wisps.
Golden Sparkles	150	Gold stars, jump burst, and apex linger.
Poison Bubbles	125	Purple/green bubbles with occasional skull symbols.
Pixel Stars	100	Blocky 1990s arcade palette and jump starburst.
Fire Trail	175	Flames and embers intensify in fast falls; underwater becomes steam or reduced flame.
Blue Ghost Fire	200	Harmless spirit flames orbit and trail.
Snowflake Trail	150	Snow density reacts to speed; frost idle; snow landing puff.
Autumn Leaf Trail	150	Multi-color rotating leaves and landing burst.
Flower Petals	150 per theme	Cherry, rose, sunflower, or fantasy petals with slow drift.
Bubble Trail	125	Upward bubbles, underwater density, and direction-change pops.
Ocean Foam	150	Land splash becomes underwater bubble stream.
Lightning Static	200	Speed-reactive sparks and jump bolt; no full-screen flashes.
Rainbow Trail	300	Short smooth, pastel, or retro spectrum with strict fade.
RGB Pixel Trail	300	Random, cycling, or custom-palette arcade pixels.
Glitch Fragments	250	Localized pixel breakup and RGB separation.
Cosmic Dust	250	Stars, tiny planets, cosmic dust, and idle orbit.
Mini Comets	225	Short-tailed comets; denser during fast falls.
Black Hole Orbit	300	Tiny cosmetic particles orbit and absorb into a small void.
Tiny Bats	200	Brief followers scatter; idle bat may hang nearby.
Ghost Companions	250	Tiny expressive ghosts wave, circle, or dissolve.
Tiny Slimes	250	Short-lived bouncing creatures that may match the skin.
Cat Paw Prints	100	Surface prints or translucent aerial paw symbols.
Demon Footprints	125	Purple burning marks with horn/tail motifs.
Hearts	75-125	Standard, gothic, pixel, broken, or rainbow hearts.
Musical Notes	150	Notes trail and may pulse with tap sounds.
Confetti	125	Speed-reactive bursts under strict particle limits.
Candy Trail	175	Bouncing sweets that fade quickly.
Crystal Shards	175	Color variants sparkle and dissolve.
Mushroom Spores	175	Soft glow and occasional tiny mushroom icons.
Fireflies	175	Independent orbiting lights, especially strong at idle.
Cloud Trail	150	Cartoon puffs with rare recognizable shapes.
Rain Cloud Companion	250	Delayed-follow cloud with cosmetic non-accumulating rain.
Halo Light	225	Soft rays and a brief apex halo.
Dark Aura	200	Purple-black aura with rising particles; never hides the hero.
Crown Sparkles	250 / earned	Prestige crown and gold particles.
Retro Score Numbers	150	Cosmetic +10, TAP!, NICE!, WOW! text with no score meaning.

Effect	Coins	Components / environment behavior
CRT Afterimage	225	Cyan/magenta/purple arcade afterimages.
Matrix Code	200	Abstract fictional symbols, not readable messages.
Floating Runes	225	Orbiting symbols briefly connect on jump.
Moon Dust	125	Gray/purple trail and broad landing puff.
Alien Spores	200	Green/purple particles with occasional eye shapes.

Premium Complete Sets

Set	Suggested price	Included package
Velvet Hex Shadow	400	Purple slime movement, shadow-smoke idle aura, purple rune jump burst, slime landing splash.
Cosmic Traveler	500	Cosmic dust, idle orbiting stars, comet jump burst, miniature black-hole landing effect.
Frozen Spirit	400	Snowflake trail, frost aura, ice-crystal jump burst, snow landing puff.
Infernal	400	Fire trail, ember aura, flame jump burst, magma landing splash.
Retro Arcade	500	RGB pixels, CRT afterimage, pixel-star jump burst, TAP! landing text.

A complete set still occupies one character-particle slot. Seasonal particle families may include Halloween, Winter, Spring, Summer, Space, Arcade, Volcano, and Ocean; earned seasonal effects remain permanently owned after the season.

25. Tap Effect and Character Particle Engineering Handoff

Source basis: Original Expanded Bible pp. 6-17 and 40-44; current store/equipment evidence. Expands the preceding cosmetic chapters without promoting example values to locked balance.

Data-Driven Tap Effect Definition

Field	Purpose / example
id / display_name	Stable semantic ID such as tap_effect_rain plus localizable player-facing name.
price / acquisition	Hero Coin price or earned-only source; mutually consistent with purchasable/exclusive fields.
effect_scene	Scene/resource path for the immediate tap response.
supports_custom_color	Whether approved hue/palette controls are exposed.
supports_accumulation	Whether temporary terrain zones remember deposits.
rare_event_id / chance	Optional rare-event eligibility. Historical rain example 0.00001 = 0.001%; tune and simulate.
max_active_instances	Per-effect cap; historical starting example 8.
duration_seconds	Lifetime before clean removal; historical starting example 4.0 seconds.
quality / accessibility	Full/reduced/minimal/off resources plus flash, shake, sound, haptic, opacity, and remote-player behavior.
enabled / version	Safe content switch, definition version, and migration/rollback identity.

Manager and Event Architecture

Component	Responsibility
TapEffectManager	Listen for valid `tap_registered(screen_position, world_position)`, resolve the equipped definition, spawn presentation, apply environment variant, request rare eligibility, and enforce caps.
TapEffectDefinition	Own data and resource references; gameplay scripts never branch on every cosmetic ID.
TapEffectInstance	One pooled visual instance with lifetime, quality, and cleanup behavior.
TemporaryEnvironmentManager	Track bounded snow/magma/leaf/water/paint/etc. accumulation zones rather than permanent per-particle objects.
RareEffectManager	Evaluate one qualified threshold/check, persist state, issue harmless presentation, and route permanent grants idempotently.
CosmeticLoadoutManager	Validate ownership, compatibility, product eligibility, and one equipped item per slot.
CharacterParticleManager	Compose trail, idle, jump, fall, landing, aura, and environment variants from movement signals.
ParticleObjectPool	Reuse emitters, decals, followers, and temporary creatures without allocation spikes.
PerformanceSettings	Apply device tier, local/remote density, accessibility overrides, and degradation order.

Required general movement signals include `character\_jump\_started`, `character\_falling\_started`, `character\_landed`, `character\_idle\_started`, and `character\_movement\_changed(velocity)`. Movement code emits state; it never references a particular cosmetic ID.

Temporary Accumulation Zones

Snow, magma/lava, leaves, paint, water, sand, flowers, mushrooms, crystals, and similar effects must not leave one permanent physics object per particle. Instead, record bounded terrain zones and update a small number of staged visuals.

Zone field	Meaning
zone_id / effect_type	Stable location bucket and material family.
current_amount / maximum_amount	Clamped accumulation state.
last_updated_time	Fade, melt, cooling, cleanup, and merge decisions.
current_visual_stage	Empty, trace, pile/pool, large, cooling/melting, or effect-specific stage.
rare_spawn_checked	Prevents a roll for every particle and repeated rolls in one qualified pile.
source/effect owner	Supports local/remote presentation limits and cleanup without granting power.

CANDIDATE THRESHOLD Historical design checks Evil Snowman or Obsidian Golem once when a qualifying snow pile or magma pool reaches 75% capacity, using an example 3-5% chance. A failed check becomes eligible again only after the old zone fully melts/disappears and a new qualified zone forms. These are tuning candidates, not locked probabilities.

Particle Loadout, Anchors, and Compatibility

Conceptual loadout fields: `character\_id`, `skin\_id`, `particle\_effect\_id`, `tap\_effect\_id`, and `primary\_badge\_id`. Add acquisition source, content version, product eligibility, and entitlement revision in durable records.

Anchor	Typical attachment
ParticleAnchorCenter	Auras and centered orbiting components.
ParticleAnchorBack	Trails and backward-moving emission.
ParticleAnchorFeet	Landing bursts, ground contact, prints, and impact decals.
ParticleAnchorHead	Crowns, halos, clouds, and head-follow components.
ParticleAnchorTail	Tail-specific sparks, ribbons, and character-aware effects.

Character-specific effects declare compatible hero/skin IDs in shop copy; universal effects declare universal behavior. The preview must exercise horizontal movement, jump, fall, idle, landing, and full/reduced quality before purchase or equip.

Quality, Multiplayer, and Performance

Quality	Required behavior
Full	Complete trail, idle, orbiting, jump/fall, landing, environment variation, and approved secondary particles.
Reduced	Lower emission, shorter trail, fewer orbiters, simplified landing, and no secondary particles.
Minimal	One basic trail or immediate tap response; no idle aura or environment variation.
Off	Cosmetic particles/effects suppressed where permitted; gameplay cues remain independently visible.

- Prefer GPUParticles2D where supported; provide CPUParticles2D or a simplified fallback.
- Pool repeating objects, cap particle count/emission/decals/zones/temporary creatures, merge nearby zones, and stop unnecessary off-screen work.
- Local player may see full presentation. Remote players use reduced count, shorter lifetime, no persistent large objects, no screen distortion, and no cosmetic audio unless enabled.
- Expose Other Player Particles: Full | Reduced | Off, while preserving mandatory gameplay signals independently.
- Rare events remain eligible at lower quality but render with a lightweight presentation; a setting cannot change rare odds or ownership eligibility.

Recommended First Twelve Character Particles

Priority	Effect	Reason in initial mix
1	Purple Slime Droplets	Brand-aligned affordable identity.
2	Shadow Smoke	Velvet Hex thematic trail/idle language.
3	Golden Sparkles	Prestige and collection readability.
4	Pixel Stars	Low-cost arcade/retro identity.
5	Fire Trail	Strong movement-state treatment.
6	Snowflake Trail	Seasonal/environment-aware treatment.
7	Bubble Trail	Water-friendly playful motion.
8	Autumn Leaves	Seasonal landing/movement variation.
9	Tiny Bats	Characterful follower particles.
10	Rainbow Trail	Premium spectrum expression.
11	Glitch Fragments	Premium arcade/corrupted identity.
12	Cosmic Dust	Premium space/world expansion identity.

PART 5

AUDIO, INTERFACE, AND PLAYER PRESENTATION

The sensory and navigational layer that makes every tap legible, satisfying, accessible, and expressive.

26. Music, Sound, and Adaptive Audio

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Shared Musical Identity

The TAM audio identity should be playful, energetic, slightly mischievous, and endlessly replayable. The core palette combines bright electronic percussion, bouncy synth bass, plucked marimba, kalimba or pizzicato-like melody, soft arcade tones, and organic accents such as claps, snaps, wood knocks, bells, whistles, and ambience.

2 Main Menu

The main menu should present a calmer, cleaner version of the TAM identity: upbeat, welcoming, lightly mischievous, and simple enough to support login, profile, shop, and mode selection without fatigue.

3 Competitive / Solo Run Music

Competitive and solo survival music should be faster, sharper, more focused, and rhythmically driving. Use punchy electronic percussion, a driving synth bass, quick plucked melodies, soft arcade tones, and short organic accents. Intensity may build as survival time increases or a match becomes close, but the track must not obscure tap timing or sound cues.

4 Infinity Taps Music

CONFIRMED Infinity Taps should use biome-specific music while preserving a shared core motif.

A practical adaptive structure uses one recognizable theme plus biome layers. Forest may introduce wood, birds, and softer percussion; snow may use bells and airy textures; beach may add water and light hand percussion; desert may add dry shakers and sparse plucks; space may thin the rhythm and widen the ambience.

5 Zone-Based Music

The approximately two-minute zone length allows the player to hear most of a track or arrangement. Music should establish the biome, expand during exploration, and transition during the final portion. Crossfades and synchronized stems are preferable to abrupt restarts.

6 Interactive Sound

- Water taps produce splash size and pitch variation.
- Bushes, leaves, wood, stone, snow, sand, mud, and metal use distinct tactile sound families.
- Animals have layered first-tap, repeated-tap, and rare-event sounds.
- Heroes have identity stingers or short reactive motifs.
- Rare effects such as lightning, snowman emergence, golem awakening, or alien contact receive memorable one-shot cues.
- Tap audio should use variation pools to avoid machine-gun repetition.

7 Current Asset Footprint and Compression

CONFIRMED Eight songs currently total approximately 27 MB.

Instruction	Developer direction
Preserve masters	Keep original WAV or FLAC masters outside the shipped build.
Primary export	Use Ogg Opus, stereo, 48 kHz, VBR. Default around 96 kbps.
Bitrate by track	Calm tracks: approximately 80-96 kbps. Energetic tracks: approximately 96-112 kbps.
Fallback	Use Ogg Vorbis quality 4-5 if the engine or loop workflow does not support Opus reliably.
Memory behavior	Stream long tracks or keep compressed in memory; do not decompress every full song on load.
Loop quality	Preserve exact loop points, trim unnecessary silence, and test gapless transitions on device.
Avoid	Do not repeatedly re-encode, drop to 22.05 kHz, or force mono merely to save a small amount.
Target	Reduce the current 27 MB music set to roughly 10-15 MB without noticeable in-game quality loss.

8 Audio Accessibility

- Independent music, sound-effect, ambience, and interface volume controls.
- Visual substitutes for important competitive audio cues.
- Reduced-intensity option for sharp or startling rare-event sounds.
- No gameplay advantage from listening at high volume.

27. Interface, Onboarding, Profiles, and Social Presentation

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Competitive Onboarding

1. Brand intro and account or guest entry.
2. A short interactive lesson: tap to rise, stop tapping to fall, avoid the ground.
3. Introduce Green Apple and Purple Fruit only after the player understands flight.
4. Complete a solo run and establish an initial performance profile.
5. Present private play, public matchmaking, profile, achievements, shop, and settings.

2 Infinity Onboarding

1. Place the hero in a small living environment immediately.
2. Prompt one tap near the hero, one tap on water, and one tap on an animal or plant.
3. Show the lifetime tap counter and next Hero Coin threshold.
4. Reveal the journal only after the player triggers a first discovery.
5. Avoid presenting failure language, ranked systems, or a dense objective list.

3 Main Navigation

- Play / Solo / Competitive or equivalent product-specific entry.
- Heroes and Collection.
- Inventory / Loadout.
- Shop.
- Achievements and Challenges.
- Profile.
- Journal in Infinity Taps.
- Settings, accessibility, account, terms, and support.

LEGACY Earlier UI planning also referenced Explore, Wiki, landing page, guest play, and 404/fallback screens. These remain useful product surfaces, but exact navigation should follow the separate-title architecture.

4 Profile

- Display name, primary badge, up to three showcase badges, profile frame, and equipped hero.
- Competitive statistics: rating, tier, season best, wins, survival records, and recent form.
- Infinity statistics: lifetime taps, distance, biomes, behaviors, heroes encountered, and rare discoveries.
- Collection provenance: earned in Infinity, ranked reward, season, achievement, shop, or supporter bundle.
- Privacy controls for recent matches, detailed statistics, and social discovery.

5 Match Presentation

Player introductions, match results, leaderboards, and opponent cards should use badges, frames, hero variants, entrances, and titles to create identity. Presentation must remain brief enough to support rapid rematches.

6 Purchase Presentation

When a hero is found in Infinity, the first tap should not instantly launch a full-screen shop. Use a small card and require a deliberate second action. Purchases must show price, ownership, source, permanence, and whether the item is available elsewhere.

PART 6

TECHNOLOGY, TRUST, AND MOBILE QUALITY

The shared account, authoritative backend, security, analytics, moderation, accessibility, performance, battery, and network foundations.

28. Accounts, Multiplayer, Backend, and Security

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Current Technical Foundation

The current project is built in Godot with gameplay and support scripts covering analytics, authentication, badges, bots, catalog, challenges, configuration, networking, online matches, pickups, players, stats, save services, Supabase authentication, and UI.

Component	Current direction / state
Client	Godot mobile-first client with gameplay, bot, catalog, achievement, save, auth, and network scripts.
Authentication	Supabase account system with email verification and planned Google OAuth.
Backend	Cloudflare Worker service at tap-against-me.dontwatchmedia.workers.dev.
Database	Cloudflare D1 production database initialized with expected application tables.
Multiplayer	Lobby and room code exists; production validation between separate devices/accounts is the next major proof.
Achievements	Shared achievement definitions; backend health currently reports 55.
Testing	53 integration tests were reported passing after backend validation.

2 Production Security State

The production backend was migrated away from the earlier hand-rolled or HS256-oriented verification path toward Supabase JWKS validation using the current EC/ES256 signing key. The one-shot schema bootstrap was completed and then disabled.

- Production D1 schema initialized.
- ADMIN_KEY removed from Cloudflare secrets after bootstrap.
- POST /v1/schema returns 404 in production.
- Preview URLs disabled.
- Production secrets limited to required Supabase configuration.
- Badge loadouts are sanitized so users can display only server-known earned badges.

3 Immediate Engineering Milestones

1. Connect the Godot client to the production Supabase project and Worker base URL through environment-safe configuration.
2. Complete real email signup, confirmation, login, refresh, logout, and recovery flows.
3. Verify automatic D1 user and profile provisioning after authenticated access.
4. Configure and validate Google OAuth on mobile.
5. Run production private matches between two physical phones using two distinct accounts.
6. Validate reconnect, timeout, result submission, duplicate submission, and version mismatch behavior.
7. Integrate shared account and catalog logic that supports two player-facing games without combining product-specific progress.

4 Multiplayer Authority

WORKING The server should be authoritative for match identity, results, rewards, rating updates, and impossible-state validation. The client may predict movement for responsiveness.

- Use deterministic or replayable simulation inputs where feasible.
- Assign client-generated idempotency keys to result submissions.
- Store physics and ruleset version with match records.
- Reject duplicate rewards and duplicate match writes.
- Preserve enough telemetry to investigate disputes without storing unnecessary personal data.

5 Shared Account Across Two Titles

The backend should represent product source explicitly. A shared user owns a hero once, but ranked rating belongs to Competitive TAM and biome progress belongs to Infinity Taps. Achievement events should include source and ruleset version so cross-title goals remain auditable.

29. Fair Play, Anti-Cheat, Moderation, and Analytics

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Anti-Cheat Philosophy

TAM should protect meaningful status without treating skilled or high-frequency tapping as suspicious by default. Detection should combine server authority, input plausibility, replay analysis, device behavior, repeated patterns, and economy anomalies.

2 Signals

- Impossible tap intervals sustained beyond human or hardware limits.
- Perfectly periodic taps with no contextual adjustment.
- Movement traces inconsistent with submitted inputs.
- Impossible pickup reaction timing.
- Repeated disconnects only when losing.
- Duplicate or reordered result submissions.
- Currency gain inconsistent with achievements, taps, or matches.
- Bot or emulator behavior used to automate Infinity tap rewards.

3 Enforcement

- Use flags and review thresholds before permanent action.
- Separate competitive restrictions from account-wide collection where proportionate.
- Explain when a match is invalidated or rating is withheld.
- Allow appeals for severe actions.
- Never sell a shortcut around enforcement or verification.

4 Moderation Scope

Social features should begin conservatively. Display names, profile text, and emotes require filtering and reporting. Direct chat is not necessary for the core game and would create moderation burden. Friend invites and preset emotes can provide social connection with lower risk.

5 Analytics

Area	Key measures
Competitive feel	Run duration distribution, death height, recovery rate, gravity phase, rematch rate
Matchmaking	Queue time, skill gap, survival gap, bot share, disconnect rate, upset rate
Pickups	Spawn-to-collect time, contest rate, benefit magnitude, bot reaction delay
Infinity engagement	Session length, taps per session, interaction diversity, biome dwell, journal completion
Living world	First-tap rate, repeated-tap rate, chain completion, rare-event discovery
Economy	Coin earn/spend, item conversion, purchase regret/refund, exclusive source integrity
Audio	Track completion, zone transition exits, download size, memory use, mute rates

Analytics should answer design questions, not merely maximize time spent. Infinity Taps in particular should not be optimized into compulsive pressure that contradicts its calming purpose.

30. Monetization and Advertising Guardrails

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

CONFIRMED Tap Against Me must not be pay-to-win. Purchases are cosmetic or presentation-focused.

1 Current Product Offers

Offer	Price	Contents / rule
Hero Coin Starter Pack	\$0.99	A small beginner currency package to help purchase an early cosmetic.
Ad-Free Pass	\$2.99	Removes ads for 30 days; not a permanent removal.
Seasonal Pass	\$4.99	Season-specific progression with cosmetics unlockable only during that season.
Supporter Bundle	\$9.99	Character bundle, character particle effect, tap effect, and 60 days without ads.

2 Prohibited Purchases

- Damage, life, regeneration, gravity resistance, tap strength, movement speed, pickup magnetism, or stamina that affects fair competition.
- Improved matchmaking, protected rank, purchased rating, or reduced rating loss.
- Higher valid tap allowance or anti-cheat exemptions.
- Infinity-exclusive earnable rewards.
- Required readability or accessibility features.

3 Advertising

WORKING Ads should never interrupt an active competitive match or a rare Infinity encounter.

- Place ads at natural session boundaries such as post-match or voluntary reward opportunities.
- Do not use deceptive close buttons or accidental purchases.
- Respect the 30-day ad-free entitlement across eligible shared products if technically and economically intended; this requires a final entitlement decision.
- Rewarded ads may offer modest cosmetic currency or convenience, never ranked power.

4 Seasonal Ethics

The seasonal pass may use exclusive cosmetics, but the player must see the season end date, reward track, required effort, and whether items can ever return. Competitive season rewards should remain earnable through play and separate from paid pass ownership.

31. Monetization and Hero Coin Economy Proposal Reference

Source basis: Original Expanded Bible pp. 83-97; consolidated monetization guardrails. All prices and earn rates below are proposals requiring economy simulation, legal/platform review, catalog-depth validation, and explicit approval.

LOCKED Money may change appearance, collection, profile expression, ad entitlement, or supporter recognition. It may never change gravity, input, jump force, recovery, lives, item odds, achievement speed, ranking, visibility, abilities, or competitive outcome.

Direct Product Concepts

Offer	Candidate price	Proposed contents / rule
One-time Hero Starter Pack	\$0.99	200 Hero Coins, once per account; later proposal supersedes an earlier 150-Hero-Coin example.
30-Day Ad-Free Pass	\$2.99	Time-limited manually repurchased entitlement; exact ad policy remains open; trusted timestamp and restore required.
Season Pass	\$4.99 / ~3 months	Free and premium cosmetic/profile tracks; no gameplay power; unlocked items remain permanent; expiry clearly communicated.
Supporter Bundle	\$9.99	Coordinated hero/skin, particle, tap effect, supporter badge, 60 days ad-free, and optional 100-Hero-Coin bonus; permanent cosmetics, expiring ad-free.

OPEN AD POLICY The no-coercion boundary is locked, but whether non-optional advertisements ship at all remains a product decision. If ads exist, placements, frequency, minors/consent, rewarded flows, ad-free coverage, offline behavior, and regional policy require separate approval.

Candidate Repeatable Hero Coin Ladder

Package	Price	Hero Coins	Approx. bonus	Cost / 100
Pocket	\$0.99	100	Base	\$0.99
Pouch	\$4.99	600	20%	\$0.83
Chest	\$9.99	1,300	30%	\$0.77
Vault	\$24.99	3,500	40%	\$0.71
Treasury	\$99.99	15,000	50%	\$0.67

CATALOG DEPTH GATE Do not launch the \$99.99 Treasury until the shop contains enough desirable content for 15,000 Hero Coins to fund several coherent collections. Do not label it Best Value or push it aggressively; \$9.99 or \$24.99 is the more credible recommended tier.

Candidate Cosmetic Price Bands

Cosmetic family	Proposed Hero Coin band	Acquisition note
Basic color variation	50-100	Affordable customization.
Basic tap effect	100-200	Immediate response with modest complexity.
Premium tap effect	250-400	Customization, accumulation, interaction, or elaborate presentation.
Basic character particle	200-350	Trail or limited multi-state package.
Premium character particle	400-600	Complete trail/idle/jump/landing package.
Standard hero skin	400-700	Full production-quality variant.
Premium animated skin	800-1,200	Material/animation complexity beyond a standard swap.
Character cosmetic bundle	1,200-2,000	Coordinated skin and effects with transparent itemized grants.
Premium themed collection	2,000-3,500	Multiple coordinated items; no exclusive earned-only content.

Candidate Free-Earning Targets

Activity	Proposed Hero Coins	Pacing intention
One casual session	10-25	A normal play session visibly advances an affordable item.
Active day	50-100	Free play remains meaningful beside the \$0.99/100 base rate.
Completed daily challenges	15-30	Combined daily assignments and bonus.
Completed weekly challenges	75-150	Larger cadence reward without mandatory daily attendance.
Average achievement	5-25	Proportional to ordinary effort and rarity.
Major achievement	25-100	Recognizes difficult or long-term accomplishment.

The historical pacing interpretation is roughly one to four active days for a basic effect, one to two weeks for a standard skin, and two to four weeks for a premium animated skin. Treat these as hypotheses to simulate against the actual 60 achievements, challenge rotation, one-coin-per-1,000-Infinity-taps rule, catalog prices, retention, and fraud controls.

Acquisition Classes and Transaction Safeguards

- Normal shop cosmetics may be earned or purchased with Hero Coins; paid and earned currency may share a wallet only if regional/legal accounting remains correct.
- Achievement cosmetics, Infinity exclusives, ranked/season rewards, and other designated earned items never become ordinary coin purchases unless the exclusivity promise is explicitly retired before release.
- Seasonal and supporter item source remains inspectable after the entitlement window; already unlocked cosmetics are permanent.
- Every platform purchase is validated, restorable where applicable, idempotent, and distinguished as consumable, non-consumable, subscription, or time entitlement.
- Ad-free expiration uses trusted time, not only the phone clock. Cancellation, failed validation, refund/reversal, reinstall, and account merge are defined and logged.
- Every Hero Coin transaction records source, reason, signed amount, catalog/config version, idempotency key, resulting balance, and reversal link where applicable.

32. Accessibility, Performance, and Mobile Quality

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Input Accessibility

- Support one-thumb play and configurable tap regions.
- Offer left-handed and right-handed UI layouts.
- Allow reduced movement sensitivity or larger directional zones.
- Provide optional haptics strength and complete haptics disable.
- Avoid requiring sustained high tap rates for all content; use alternate goals and pacing.

2 Visual Accessibility

- High-contrast ground and collision boundaries.
- Color-independent indicators for beneficial and harmful fruit.
- Reduced flashing and reduced particle-density modes.
- Text scaling and readable mobile font sizes.
- Effect preview warnings for rapid flashes or screen shake.

3 Motion and Sensory Settings

- Camera shake strength.
- Screen flash intensity.
- Particle density.
- Background motion.
- Rare-event intensity.
- Music, ambience, effects, UI, and haptics controls.

4 Performance Targets

The game should target smooth, consistent physics on a broad range of Android devices. Competitive simulation quality is more important than maximum particle density. Use object pooling, effect caps, streamed music, compressed textures, deterministic cleanup, and scalable environmental population.

5 Network Quality

- Show connection state without distracting from play.
- Use prediction and reconciliation that do not create unexplained ground deaths.
- Define timeout and reconnect rules before ranked launch.
- Record latency and version data with match results.
- Avoid matching across unacceptable latency merely to reduce queue time.

6 Battery and Data

Infinity Taps may be used for long fidget sessions. It needs low-power options, frame-rate caps, reduced ambience, and predictable data use. Downloadable biome music or optional asset packs can be considered later, but the base experience should remain complete.

33. Mobile Viewport, Session, and Interaction Baselines

Source basis: TAM Game Bible and cross-source mobile/accessibility guidance. Values are working planning targets, not a promise that players must stay for a minimum duration.

Baseline	Working value	Design consequence
Reference viewport	360 x 640 portrait	Core gameplay, HUD, dialogs, safe areas, and touch targets remain usable at this baseline before adapting upward.
Quick survival session	15 seconds-3 minutes	Fast restart, low navigation cost, immediate input, concise result.
Ranked set	3-10 minutes	Queue, intros, rematches, and result progression respect short mobile availability.
Infinity session	30 seconds-30+ minutes	Equally valid as a brief regulation/fidget break or long exploration session; no stopping penalty.
Collection/store session	1-5 minutes	Fast preview, compare, acquisition source, equip, and return to prior context.
Season/challenge check	1-3 minutes	Clear progress and claim state without forcing play or fear-of-missing-out loops.

Test the viewport across cutouts, gesture/navigation bars, tall/narrow phones, tablets, foldables, text scaling, localization expansion, and one-handed reach. The 360 x 640 baseline is a layout floor, not permission to hard-code coordinates or crop wider/taller devices.

Mobile Interruption Contract

- Incoming calls, notifications, audio-route changes, lock/unlock, split-screen, backgrounding, low-memory recreation, and network handoff produce an explainable pause, reconnect, resume, or result.
- Infinity saves before/while backgrounding and returns without scolding, timer loss, or missed exclusive opportunity where pending state can be preserved.
- Competitive play cannot continue accepting hidden touches under overlays; reconnect/abandonment policy distinguishes technical interruption from repeated intentional quitting.
- Music and ambience resume/crossfade safely; short effects do not replay as duplicate grants or startling bursts after resume.
- Quality tier may reduce spectacle under thermal/memory pressure but never remove required gameplay cues or change simulation.

PART 7

DELIVERY, LIVE OPERATIONS, AND DECISIONS

MVP boundaries, release gates, phased development, content cadence, risks, canonical decisions, and open questions.

34. MVP Definition and Development Roadmap

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Scope Principle

The project is already deep in networking and multiplayer work. The correct move is not to discard that foundation. Competitive TAM is closer to a playable proof; Infinity Taps needs substantially more content. Development should finish the current networking milestone while prototyping Infinity at deliberately small scale.

2 Competitive TAM MVP

Area	MVP requirement
Core gameplay	Reliable tap-to-rise physics, gravity escalation, ground failure, restart, and records.
Modes	Solo survival and private online 1v1; public queue may use bot backfill.
Items	Green Apple, Purple Fruit, and controlled Hero Coin pickup behavior.
Bots	At least several imperfect difficulty profiles using the same rules.
Accounts	Email account flow, Google login, distinct profiles, cloud persistence.
Multiplayer	Two physical phones, two accounts, match discovery/invite, synchronized play, results.
Progress	Initial achievements, badges, Hero Coin wallet, cosmetic loadout.
Presentation	Main menu, profile, shop, settings, results, rematch.
Fairness	Authoritative result validation, duplicate protection, basic anti-cheat flags.
Audio	Menu and gameplay music, tap and physics cues, compressed asset pipeline.

3 Infinity Taps MVP

Area	MVP requirement
Environment	One polished biome or zen playground.
Movement	Tap-directed traversal with no loss condition.
Reactivity	Water, bush, tree/leaves, ground, and at least one additional object.
Animals	One or two animals with multi-stage reactions.
Progress	Persistent session and lifetime tap counts; one Hero Coin per 1,000 valid taps.
Discovery	Small journal, a few behaviors, several achievements.
Hero content	One discoverable hero or variant and one hidden Infinity-exclusive reward.
Audio	One adaptive biome track or layered arrangement plus environmental sound.
Persistence	Resume safely without losing discoveries or rare encounter state.

4 Recommended Phases

Phase	Timing	Outcome
Phase 0 - Foundation	Current / substantially complete	Godot project, backend, database, auth foundation, achievement definitions, tests, deployment security
Phase 1 - Competitive Vertical Slice	Near term	Polish solo physics, items, UI, bot feel, records, one gameplay track
Phase 2 - Real Account and Private Match	Near term	Email and Google login, two-device match, production result persistence
Phase 3 - Public Queue	Short term	Matchmaking, bot backfill, ghost experiments, queue UX, basic anti-cheat
Phase 4 - Ranked Seasons	Medium term	Glicko, tiers, 90-day season, rewards, leaderboards, human-only upper-tier policy
Infinity Phase 1 - Zen Playground	Parallel small prototype	One biome, living reactions, animals, tap counter, hidden reward
Infinity Phase 2 - Biome Journey	After core proof	Two-minute zones, multiple biomes, boosts, adaptive music, journal expansion
Infinity Phase 3 - Infinite Adventure	Long term	Animals, heroes, chains, hidden events, seasonal worlds, deep ocean, planets, rare encounters

5 Release Gate

Do not launch ranked merely because online matches connect. Ranked requires stable physics versioning, credible matchmaking, explainable results, disconnect rules, reward integrity, and sufficient bot disclosure. Do not launch Infinity with many empty biomes; one deeply reactive biome is more faithful than ten decorative backgrounds.

35. Content Production and Live Operations

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Content Units

Unit	Minimum definition
Hero	Silhouette, movement, idle, reactions, lore line, ownership record, loadout compatibility
Variant	Distinct visual/theme, source, encounter or purchase rule, journal entry
Biome	Visual set, collision set, ambient population, interaction set, music layer, transition
Animal	Idle behavior, navigation, first tap, repeated tap, rare response, cooldown
Tap effect	Tap response, persistence, rare event, cleanup, preview, competitive visibility rating
Particle	Hero attachment, movement response, density levels, preview, performance budget
Achievement	ID, title, description, event source, threshold, tier, reward, secret flag
Season	Theme, dates, ranked rules, reward table, pass track, event content, end-state handling

2 Seasonal Cadence

Competitive ranked seasons last 90 days. Each season should offer a distinct cosmetic identity and may introduce a controlled new feature, hero, effect, arena, or world event. The season must not destabilize core physics without sufficient unranked testing.

3 Infinity Seasonal Worlds

Infinity can reflect seasons through weather, animal behavior, music layers, heroes, and rare events. Seasonal worlds should enrich the living environment rather than simply recolor it. Permanent journal history should record that the player witnessed the event.

4 Content Reuse

- A hero created for ranked presentation can later receive Infinity encounter behavior.
- A tap effect created for the shop can gain biome-specific interactions.
- A music motif can be rearranged for menu, competitive, and biome use.
- A seasonal cosmetic may have different unlock paths per title, but ownership must remain unambiguous.

5 Quality Bar

Content should be judged by response depth, readability, personality, and source clarity. A new animal with one looping animation is less valuable than an existing rabbit with seven discoverable behaviors. A new effect that hides the ground is not acceptable merely because it looks impressive.

36. Risks, Scope Controls, and Design Safeguards

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

Risk	Failure mode	Safeguard
Two products become unrelated projects	Duplicate systems, inconsistent ownership, slower delivery	Share accounts, catalog schema, hero IDs, effects, achievements framework, and audio motif; separate player-facing loops and progression.
Infinity becomes a background viewer	World feels dead after novelty fades	Require multi-stage reactions, chains, ambient behavior, and journal depth before adding more biomes.
Competitive population is too small	Long queues or unfair matches	Bots, ghosts, private matches, solo records, gradual search expansion, and transparent AI labeling.
Bots feel unfair	Players distrust results	Same rules, reaction delay, imperfect perception, no future knowledge, telemetry-driven tuning.
Monetization weakens prestige	Exclusive rewards lose meaning	Protect Infinity earn-only and ranked rewards; show item source.
Effects damage readability	Competitive deaths feel unfair	Effect visibility rating, reduced-particle option, collision overlays, ranked-safe whitelist.
Achievement catalog drifts	Client and server disagree	Single shared definitions, versioned events, automated tests, export authoritative catalog.
Music size grows rapidly	Large app and memory footprint	Opus/Vorbis import preset, stream tracks, reuse stems, target bitrate, loop QA.
Networking milestone is abandoned	Existing investment is wasted	Finish two-device private-match proof before broad content expansion.
Scope expands before feel is proven	Many systems, weak core	Vertical slices and release gates; one deep biome before many shallow biomes.

1 Non-Negotiable Safeguards

- No pay-to-win.
- No traditional failure in Infinity Taps.
- No hidden bot impersonation.
- No purchasable Infinity-exclusive earnables.
- No unreadable competitive ground or collision state.
- No ranking launch before result and matchmaking integrity.
- No biome expansion that outruns interaction depth.

37. Canonical Decision Register and Open Questions

Source basis: TAM Game Design Bible v1.0 primary canonical narrative, synthesized from the full supplied Bible set.

1 Canonical Decisions

- Official slogan: Every Tap Against Me Takes You Further.
- Android-first, portrait-oriented, mobile-friendly 2D design.
- Competitive TAM centers on tapping to remain airborne; ground contact ends the standard survival round.
- Infinity Taps is noncompetitive, no-loss, fidget and exploration-focused.
- Current product direction: separate player-facing titles under one shared TAM ecosystem.
- Hero is the current canonical character term.
- Hero Coin is the canonical currency term; golden token may be its pickup representation.
- Infinity Taps awards one Hero Coin per 1,000 valid taps and does not normally spawn coins.
- Infinity-earned designated cosmetics and variants cannot be purchased.
- Animals and environment have first-tap and repeated-tap reactions.
- Found unowned heroes or variants can present purchase or unlock opportunities in-world.
- Zones target approximately two minutes.
- Infinity boosts include velocity, jet engine hold, anti-gravity, and extra-bouncy.
- Competitive matchmaking uses Glicko-style rating plus rolling survival performance.
- Visible tiers include Bronze, Silver, Gold, Platinum, and future upper tiers.
- Ranked seasons reset every 90 days.
- Bots are acceptable for population support but use the same rules and imperfect behavior.
- One primary badge plus up to three showcase badges.
- Monetization is cosmetic-only with the established \$0.99, \$2.99, \$4.99, and \$9.99 offers.
- Eight current music tracks total about 27 MB and should be shipped through a compressed streaming pipeline.

2 Open Product Questions

1. Final public naming of the Infinity product and whether the subtitle "A Tap Against Me Adventure" is used.
2. Whether Hero Coin is one wallet across both titles or has title-specific earning caps and entitlements.
3. Exact ranked tier names above Platinum, division count, rank floors, and human-only cutoff.
4. Exact reset compression formula for 90-day seasons.
5. Whether First to Zero and Most Taps become permanent modes, events, or are removed.
6. Final horizontal-control model for competitive taps.
7. Exact Green Apple and Purple Fruit values, durations, and spawn rules.
8. Which Infinity boosts are universal versus biome-specific.
9. Whether encounter-exclusive purchases remain available after discovery or only during the encounter.
10. Whether 30-day ad-free and 60-day supporter entitlements apply across both products.
11. Complete reconciliation of the 55 implemented achievements with final design names, thresholds, and rewards.
12. Final art style, hero proportions, and animation budget.
13. Whether optional controller support remains a future platform feature.

3 Recommended Next Decision Workshop

The next design lock should focus on four items: competitive physics values, two-device private-match acceptance criteria, shared-account data boundaries, and the one-biome Infinity vertical slice. Those decisions unlock engineering and content production without requiring every future biome, hero, or season to be specified.

PART 8

DETAILED REFERENCE APPENDICES

Traceable catalogs, implementation snapshots, data contracts, release evidence, unresolved decisions, legacy context, and source custody.

38. Appendix A. Terminology

Source basis: TAM Game Design Bible v1.0 canonical appendix.

Term	Definition
TAM	Tap Against Me; the competitive flagship and shared universe shorthand.
Infinity Taps	The no-loss exploration and fidget title within the TAM ecosystem.
Hero	Playable or collectible character.
Variant	A distinct version of a hero with separate source or ownership.
Hero Coin	Cosmetic currency; not competitive power.
Golden Hero Token	Earlier term; recommended as the visible pickup form of Hero Coin.
Valid Tap	A tap accepted by the active gameplay rules and input window.
Survival Time	Duration before ground contact or elimination.
Glicko	Rating family that tracks rating and uncertainty.
Performance Profile	Rolling survival and behavior metrics used alongside rating.
Primary Badge	The single badge displayed beside the username.
Showcase Badges	Up to three additional profile badges.
Zone	An approximately two-minute Infinity travel segment.
Biome	A themed environment with unique visuals, audio, objects, animals, and reactions.
Interaction Chain	A tap response that causes another object or character to react.
Infinity-exclusive	An item that must be earned through Infinity Taps and cannot be purchased.
Ranked-safe	A cosmetic or modifier that preserves clarity, symmetry, and competitive integrity.

39. Appendix B. Achievement Seed Catalog

Source basis: TAM Game Design Bible v1.0 canonical appendix.

The following names were explicitly discussed. The deployed backend currently reports 55 definitions; this appendix is the recovered design-name set and must be reconciled with code for complete thresholds and rewards.

Fruit and Poison

- Life of Poisonous Fruit
- Anti-Gravity
- Fruit Salad
- Toxic Appetite
- Poison-Proof
- Apple a Day
- Orchard Raider
- Clean Eating
- No Boost Needed
- Bad Decisions

Survival and Control

- First Flight
- Learning to Fly
- Gravity Defier
- Sky Eternal
- Terminal Velocity
- Close Call
- Living on the Edge
- Perfect Control
- Untouchable
- One More Tap

Coins and Shop

- Coin Collector
- Golden Hoard
- Token Hunter
- Big Spender
- Shop Regular
- Fully Dressed

Multiplayer and Progression

- First Victory
- Getting Started

Secret

- Was That Worth It?
- Floor Inspector
- Maximum Poison
- Perfectly Balanced
- Velvet's Nine Lives
- Too Greedy

Infinity Exploration

- World Traveler
- Into the Clouds
- Beyond the Sky
- One Small Tap
- Going Deep
- At the Bottom
- First Contact

CATALOG RECONCILIATION REQUIREMENT Export achievement ID, title, description, event source, threshold, tier, reward, secret flag, and product scope from the authoritative shared definitions. Resolve duplicate or incomplete historical names before release.

40. Appendix D. Developer Acceptance Criteria

Source basis: TAM Game Design Bible v1.0 canonical appendix.

D.1 Competitive Core

- A player can begin, tap to rise, fall under gravity, collide with the visible ground, receive an accurate result, and restart without state corruption.
- Physics behavior is consistent at supported frame rates and recorded with a ruleset version.
- Green Apple and Purple Fruit effects are readable, configurable, and logged.
- A bot uses the same physics and cannot react to an item before a human-readable spawn window.

D.2 Two-Device Private Match

- Two physical Android phones sign in with different accounts.
- One account can find, invite, or match with the other.
- Both clients enter the same authoritative match.
- Tap inputs and movement remain acceptably synchronized under normal latency.
- Ground contact and winner agree on both clients and the server.
- The result writes once, rewards once, and updates both profiles.
- Disconnect, timeout, duplicate submission, and rematch paths are tested.

D.3 Infinity Vertical Slice

- No input or environmental event creates a loss screen.
- Water splashes, a bush wiggles, tree leaves fall, and reactions vary with repeated taps.
- At least one animal has three distinct interaction stages.
- A found hero or variant displays an in-world identification and acquisition path.
- Lifetime taps persist; one Hero Coin is awarded at the 1,000-valid-tap threshold.
- One Infinity-exclusive reward is earnable and absent from direct purchase catalogs.
- The biome track transitions or loops cleanly and environmental sounds vary.

D.4 Economy and Entitlements

- The server rejects duplicate reward transactions.
- Badge loadouts accept only earned badges.
- Infinity-exclusive source flags block shop purchase.
- Ad-free expiration is stored and interpreted consistently.
- No cosmetic changes simulation values in ranked rulesets.

D.5 Audio Import

- Masters remain lossless and outside the shipping asset set.
- Shipped music uses the agreed Opus or Vorbis preset.
- Long tracks stream and do not cause load spikes.
- Loop points are tested on target Android hardware.
- The eight-track set is reduced toward the 10-15 MB target without obvious artifacts.

41. Appendix E. Legacy and Reserved Concepts

Source basis: TAM Game Design Bible v1.0 canonical appendix.

Concept	Historical idea	Current treatment
Puffs	Earlier character or companion language with passive bonuses, artifact discovery, currency, abilities, rarity, and profile customization.	Reserve as companion or alternate creature class; do not replace current Hero terminology without a deliberate change.
Artifacts	Collectible objects tied to exploration, buffs, rarity, or journal progression.	Could return as non-power lore collectibles or event objects; avoid ranked stat advantage.
Bubble / moving targets	Earlier active-tapping target loop with accuracy and unpredictable targets.	Possible event mode or Infinity interaction, not the current airborne core.
Stamina / cooldown	Earlier tapping-system consideration.	Only reintroduce if needed for feel or accessibility; avoid making the core feel artificially throttled.
MMO systems	Rankings, guilds, shared events, trading, cooperative and competitive goals.	Long-term universe possibilities; trading and guilds are well outside current scope.
Controller support	Optional controller input was mentioned.	Future accessibility or platform expansion; mobile touch remains primary.
Life / regeneration upgrades	Earlier mode concepts included life, regeneration per tap, and total life.	May support a specific alternate mode but cannot be purchasable ranked power.

FINAL PRODUCT STATEMENT Tap Against Me is a shared universe where tapping can mean survival, rivalry, expression, relaxation, or discovery. Competitive TAM asks how long and how well the player can resist the ground and another challenger. Infinity Taps asks how far curiosity can carry them through a world that notices every touch. In both, every tap takes the player further.

42. Feature Status, Phase, and Next-Gate Ledger

Source basis: Canonical intent from all supplied bibles; implementation evidence from the supplied implementation snapshots and current workspace catalogs. Runtime production status still requires a build/backend audit.

READ THIS FIRST The design-state and evidence-state columns are independent. 'LOCKED / PLANNED' means the requirement is firm but not demonstrated as built. 'WORKING / IMPLEMENTED' means code may exist while the intended rule is still being refined. This ledger is the fastest reference for what should happen, what evidence exists, and what must happen next.

Feature	Exact intent	Design state	Evidence state	Phase and next gate
Tap input and airborne resistance	Tap applies readable upward resistance against a fall; ground contact ends Competitive runs.	LOCKED	IMPLEMENTED baseline	Phase 0: validate feel, frame-rate invariance, input filtering, deterministic replay.
Rising gravity	Difficulty increases through authoritative, configurable gravity rather than sky darkness.	LOCKED / TUNING	IMPLEMENTED baseline	Phase 0: approve curve/caps and pin physics version.
Green Apple	Beneficial pickup with clear spawn, collection, sound, visual, and result tracking.	CONFIRMED / TUNING	IMPLEMENTED baseline	Phase 0: verify effects, bot parity, network authority, accessibility cues.
Poison Fruit	Readable gravity/debuff pickup with explicit duration, stacking, cleanse, and result evidence.	CONFIRMED / TUNING	IMPLEMENTED baseline	Phase 0: resolve final modifier and stacking matrix.
Golden Hero Token	High-value collectible/reward with distinct sunset-safe visual language.	CONFIRMED / TUNING	IMPLEMENTED baseline	Phase 0/3: validate grant ledger and rename legacy token language where necessary.
Solo survival	Rapid-start, rapid-retry competitive ruleset used to prove the core feel.	CONFIRMED	IMPLEMENTED baseline	Phase 0: complete result, record, interruption, and device-quality gates.
Private two-device match	Two accounts complete the same authoritative match with synchronized rules and results.	CONFIRMED	UNKNOWN / AUDIT	Phase 1: prove latency, reconnect, duplicate submission, and progression sync.
Public matchmaking	Find a compatible human, expand predictably, then use disclosed bot backfill where allowed.	CONFIRMED	PLANNED / AUDIT	Phase 4: population test, queue policy, regions, latency ceiling, bot thresholds.
Glicko-2 ranked	Underlying rating plus readable tier/division ladder, placements, history, and server-side updates.	CONFIRMED	PLANNED	Phase 4: resolve division count and thresholds; validate reference calculations.
Ranked seasons	Ninety-day competitive seasons with soft compression and permanent history.	CONFIRMED	PLANNED	Phase 4/5: approve rewards, peak/final policy, rollover rehearsal.

Feature	Exact intent	Design state	Evidence state	Phase and next gate
Bots	Human-like population bridge using the same physics/items, delays, mistakes, personalities, and disclosure.	CONFIRMED / TUNING	PARTIAL per source	Phase 4: seeded behavior simulation, fairness review, upper-tier restrictions.
Ghost opponents	Asynchronous or replay-driven opponent presentation without pretending to be live.	CANDIDATE	NOT STARTED / AUDIT	Post-MVP: define evidence, disclosure, rating eligibility, replay determinism.
Competitive day/night	Continuous 120-second cosmetic cycle synchronized from authoritative match data and independent of gravity.	LOCKED	PLANNED / AUDIT	Dedicated phases 1-7: baseline, controller, profiles, anchors, ambience, biome reaction, optimization.
Infinity no-loss contract	Open-ended local-first tapping and directional exploration with no fail screen, punishment, or urgency.	LOCKED	PLANNED	Phase 2: one-biome vertical slice must pass every no-loss gate.
Infinity movement languages	Land, water, sky, space, and later biomes interpret direction with distinct impulse/drag/camera feel.	CONFIRMED / TUNING	PLANNED	Phase 2: prove land plus one transition; tune without reward dependency.
Infinity tap milestone	One Hero Coin per 1,000 valid lifetime taps with persistent remainder and idempotent grant.	LOCKED	PLANNED	Phase 2: 999/1000/1001 and cross-device crash/merge tests.
Living-world reactions	Objects, terrain, water, plants, animals, and heroes support stages, context, cooldown, chains, and secrets.	CONFIRMED	PLANNED	Phase 2: water, bush, tree, one three-stage animal, one chain.
Discovery Journal	Persistent biomes, landmarks, behaviors, heroes/variants, secrets, records, and partial-progress entries.	CONFIRMED	PLANNED	Phase 2: schema, entry visibility, save migration, first-biome coverage.
Rare discoveries and pity	Rare events retain wonder while drought protection prevents implausible long-term failure.	WORKING / OPEN TUNING	NOT STARTED	Simulate base odds, interval, pity, reset, secrecy, and grant idempotency.
Infinity-exclusive rewards	Designated items are earned only in Infinity, retain source labels, and are never sold anywhere.	LOCKED	PLANNED	Phase 2: one exclusive earned in slice and automated absence-from-store invariant.
Heroes and variants	Exact hero and variant identities have personality, acquisition source, ownership, encounter, and reaction state.	CONFIRMED	PARTIAL catalog/art	Phase 3: approve launch roster, exact variant IDs, preview/encounter/acquisition flows.
Shared account/cross-save	One shared identity and eligible ownership with separate Competitive and Infinity progression records.	CONFIRMED	UNKNOWN / AUDIT	Phase 1/2: guest conversion, provider links, merge, deletion, recovery, entitlements.

Feature	Exact intent	Design state	Evidence state	Phase and next gate
Hero Coin ledger	Auditable shared currency transactions; cached balance never replaces the authoritative ledger.	LOCKED	PARTIAL / AUDIT	Phase 3: idempotency, reversal, reconcile, offline pending grant, fraud tests.
Achievements	Stable versioned criteria, retroactive and online evidence, once-only rewards, badges, and hidden rules.	CONFIRMED	DATA-DEFINED: 60	Phase 3: runtime-reachability suite for all 60; correct stale 55-count references.
Badges	Earned identity display with one primary and up to three showcase badges; never gameplay power.	CONFIRMED	DATA-DEFINED: 60	Phase 3: equip/profile runtime, animation/accessibility, source mapping validation.
Daily/weekly challenges	UTC rotations with server-witnessable progress, clear online labels, and once-only grants.	CONFIRMED / TUNING	DATA-DEFINED: 12 daily, 9 weekly	Phase 3: issuance, expiry, impossible-asignment, bonus, offline/reconnect tests.
Store and equipment	Permanent ownership separated from equipped selection; previews and purchase restoration; no power sales.	LOCKED boundary	PARTIAL: 23 records	Phase 3: fix duplicate 'none' ID collision; validate every grant, bundle, restore, and slot.
Current tap-effect loadout	One equipped cosmetic tap effect: Shockring, Starburst, Ripple, or No Effect.	CONFIRMED	IMPLEMENTED / catalog evidence	Phase 3: verify persistence, preview, remote density, no-physics golden test.
Expanded tap effects	Thirty candidate effects with bounded accumulation, optional rare depth, accessibility, and performance caps.	CANDIDATE	PLANNED	Phase 3+: prioritize ten distinct low-risk effects; data-drive behavior.
Character particles/trails	One cosmetic slot may contain trail, idle, jump/fall, landing, and environment variants.	CONFIRMED boundary	PARTIAL: 3 purchasable trails	Phase 3+: validate slot semantics, 43-effect backlog, density tiers, remote-player controls.
Audio and adaptive music	Shared motif with product-specific tone, legible cues, variation, bus controls, and Opus-first delivery.	CONFIRMED	PARTIAL / AUDIT	Phase 0-5: approve motif, cue matrix, compression, streaming, interruption, comfort.
Accessibility	Text, contrast, color-independent cues, reduced motion/flash/shake, haptics, audio, remote effects, input comfort.	LOCKED boundary	UNKNOWN / AUDIT	Every phase: device/settings matrix and accessibility acceptance evidence.
Performance quality tiers	Android-first 30/60 FPS presentation tiers preserve gameplay information while reducing spectacle.	CONFIRMED / TUNING	PARTIAL / AUDIT	Define device floor and budgets; soak, thermal, battery, load, memory tests.

Feature	Exact intent	Design state	Evidence state	Phase and next gate
Monetization/ads	Cosmetic, transparent, optional, restorable; no pay-to-win or required ads; supporter/pass concepts remain policy work.	LOCKED guardrails / OPEN packaging	PARTIAL / AUDIT	Phase 3/5: legal/store review, exact offers, opt-in placements, refund/support flows.
Live operations	Seasons and events extend cosmetics/world activity without erasing ownership or destabilizing ranked physics.	CONFIRMED	PLANNED	Phase 5: capacity plan, event rollback, reruns, content versioning, permanent history.

Maintenance rule: whenever evidence changes, update the evidence state, missing scope, next gate, acceptance evidence, and linked decision together. Never mark a feature IMPLEMENTED merely because a menu entry, JSON row, or isolated class exists.

43. Current Implemented Achievement Catalog

Source basis: *Workspace data/achievements.json*. This is an implementation snapshot, not proof that every entry has passed live validation.

IMPLEMENTATION SNAPSHOT The current catalog contains 60 achievement records. IDs and tracked-stat names are implementation contracts; display copy, targets, rewards, and enablement may be migrated or tuned only through versioned data.

Retroactive achievements must evaluate durable lifetime statistics. Online-only achievements require authoritative match evidence. Hidden achievements must not leak their exact criteria through client logs, public analytics payloads, or unearned UI copy. Tier groups must grant every completed lower tier exactly once.

ID	Display name	Category	Requirement	Reward	Flags
fruit_poison_25	Poison Fruit Consumer (Bronze)	fruit	poison_fruits_consumed >= 25	3 coins; badge_poison_bronze	retro
fruit_poison_100	Poison Fruit Consumer (Silver)	fruit	poison_fruits_consumed >= 100	5 coins; badge_poison_silver	retro
fruit_poison_500	Life of Poisonous Fruit	fruit	poison_fruits_consumed >= 500	10 coins; badge_poison_gold	retro
fruit_poison_2500	Poison Fruit Consumer (Mythic)	fruit	poison_fruits_consumed >= 2500	50 coins; badge_poison_mythic	retro
fruit_apple_500	Anti-Gravity	fruit	apples_collected >= 500	10 coins; badge_golden_apple	retro
fruit_salad	Fruit Salad	fruit	>= 1	5 coins; badge_fruit_salad	standard
fruit_toxic_appetite	Toxic Appetite	fruit	round_poison_fruits >= 5	10 coins; badge_toxic_slime	standard
fruit_poison_proof	Poison-Proof	fruit	>= 1	15 coins; badge_poison_shield	standard
fruit_apple_a_day	Apple a Day	fruit	apple_collection_day_streak >= 7	15 coins; badge_apple_calendar	retro
cookie_10	Star Baker (Bronze)	cookie	space_cookies_eaten >= 10	3 coins; badge_cookie_bronze	retro
cookie_50	Star Baker (Silver)	cookie	space_cookies_eaten >= 50	6 coins; badge_cookie_silver	retro
cookie_250	Gravity Optional	cookie	space_cookies_eaten >= 250	15 coins; badge_cookie_gold	retro
cookie_1000	The Whole Jar	cookie	space_cookies_eaten >= 1000	40 coins; badge_cookie_mythic	retro
cookie_moonshot	Moonshot	cookie	round_space_cookies >= 5	20 coins; badge_cookie_flight	standard
fruit_orchard_raider	Orchard Raider	fruit	round_apples >= 25	20 coins; badge_apple_basket	standard
fruit_clean_eating	Clean Eating	fruit	>= 1	10 coins; badge_pure_apple	standard
fruit_no_boost	No Boost Needed	fruit	>= 1	15 coins; badge_natural_flyer	standard
fruit_bad_decisions	Bad Decisions	fruit	>= 1	5 coins; badge_questionable_fruit	standard

ID	Display name	Category	Requirement	Reward	Flags
survival_30	First Flight	survival	best_survival_seconds >= 30	5 coins; badge_tiny_wings	retro
survival_60	Learning to Fly	survival	best_survival_seconds >= 60	10 coins; badge_silver_wings	retro
survival_120	Gravity Defier	survival	best_survival_seconds >= 120	20 coins; badge_gravity	retro
survival_300	Sky Eternal	survival	best_survival_seconds >= 300	50 coins; badge_eternal_wings	retro
survival_terminal_velocity	Terminal Velocity	survival	seconds_after_max_gravity >= 20	25 coins; badge_falling_star	standard
survival_close_call	Close Call	survival	close_calls >= 1	5 coins; badge_scraped_paw	retro
survival_living_on_edge	Living on the Edge	survival	round_close_calls >= 5	15 coins; badge_razor_edge	standard
survival_perfect_control	Perfect Control	survival	>= 1	15 coins; badge_balance	standard
survival_untouchable	Untouchable	survival	>= 1	15 coins; badge_empty_sky	standard
survival_one_more_tap	One More Tap	survival	longest_near_ground_streak >= 10	10 coins; badge_last_tap	standard
coins_balance_500	Coin Collector	coins	highest_coin_balance >= 500	25 coins; badge_coin_collector	retro
coins_balance_2500	Golden Hoard	coins	highest_coin_balance >= 2500	50 coins; badge_golden_vault	retro
coins_token_hunter	Token Hunter	coins	gameplay_coins_collected >= 1000	25 coins; badge_token_hunter	retro
coins_big_spender	Big Spender	coins	coins_spent >= 500	15 coins; badge_shopping_bag	retro
coins_shop_regular	Shop Regular	coins	unique_shop_items_purchased >= 10	20 coins; badge_merchant	retro
coins_fully_dressed	Fully Dressed	coins	>= 1	10 coins; badge_wardrobe	standard
coins_golden_opportunity	Golden Opportunity	coins	round_hero_tokens >= 3	15 coins; badge_golden_hero	standard
coins_shiny_distraction	Shiny Distraction	coins	>= 1	5 coins; badge_dropped_coin	standard
online_win_1	First Victory	multiplayer	online_matches_won >= 1	10 coins; badge_first_crown	online, retro
online_win_10	Last One Floating	multiplayer	online_matches_won >= 10	25 coins; badge_survivor_crown	online, retro
online_win_100	Sky Champion	multiplayer	online_matches_won >= 100	100 coins; badge_champion_wings	online, retro
online_photo_finish	Photo Finish	multiplayer	>= 1	20 coins; badge_photo_finish	online
online_flawless	Flawless Victory	multiplayer	>= 1	15 coins; badge_perfect_crown	online
online_poisoned_champion	Poisoned Champion	multiplayer	>= 1	25 coins; badge_toxic_crown	online

ID	Display name	Category	Requirement	Reward	Flags
online_comeback	Comeback Flight	multiplayer	>= 1	15 coins; badge_comeback	online
online_crowded	Crowded Skies	multiplayer	>= 1	25 coins; badge_crowded_sky	online
online_rivalry	Rivalry	multiplayer	>= 1	10 coins; badge_rival	online
online_friendly	Friendly Competition	multiplayer	private_matches_completed >= 10	15 coins; badge_friends	online, retro
prog_getting_started	Getting Started	progression	achievements_completed >= 5	10 coins; badge_beginner	retro
prog_badge_collector	Badge Collector	progression	badges_unlocked >= 10	20 coins; badge_badge_case	retro
prog_achievement_hunter	Achievement Hunter	progression	achievements_completed >= 25	50 coins; badge_achievement_hunter	retro
prog_completionist	Completionist	progression	>= 1	0 coins; badge_completionist; hex_completionist	retro
prog_dedicated_tapper	Dedicated Tapper	progression	rounds_played >= 100	20 coins; badge_tapp_counter	retro
prog_tap_veteran	Tap Veteran	progression	rounds_played >= 1000	75 coins; badge_veteran	retro
prog_persistent	Persistent	progression	immediate_restarts >= 25	15 coins; badge_never_quit	retro
prog_daily_flyer	Daily Flyer	progression	daily_play_streak >= 7	20 coins; badge_weekly_wings	retro
secret_worth_it	Was That Worth It?	secret	>= 1	5 coins; badge_regret	hidden
secret_floor_inspector	Floor Inspector	secret	>= 1	0 coins; badge_floor	hidden
secret_maximum_poison	Maximum Poison	secret	round_poison_fruits >= 7	25 coins; badge_purple_skull	hidden
secret_perfectly_balanced	Perfectly Balanced	secret	>= 1	10 coins; badge_balance_fruit	hidden
secret_nine_lives	Velvet's Nine Lives	secret	>= 1	25 coins; badge_nine_lives	hidden
secret_too_greedy	Too Greedy	secret	>= 1	0 coins; badge_greedy_hero	hidden

Current Badge Definitions

DATA-DEFINED The badge catalog contains 60 records with complete one-to-one coverage. Rarity distribution: 11 common, 26 rare, 18 gold, and 5 mythic. Four definitions are animated. Badge presentation is a reward/identity layer and never a gameplay modifier.

Badge ID	Display name	Rarity	Source achievement	Glyph	Tier	Flags
badge_cookie_bronze	Bronze Cookie	common	cookie_10	cookie	bronze	standard
badge_cookie_silver	Silver Cookie	rare	cookie_50	cookie	silver	standard

Badge ID	Display name	Rarity	Source achievement	Glyph	Tier	Flags
badge_cookie_gold	Gravity Optional	gold	cookie_250	cookie	gold	standard
badge_cookie_mythic	The Whole Jar	mythic	cookie_1000	cookie	mythic	standard
badge_cookie_flight	Moonshot	gold	cookie_moonshot	cookie	none	standard
badge_poison_bronze	Bronze Poison	rare	fruit_poison_25	poison	bronze	standard
badge_poison_silver	Silver Poison	rare	fruit_poison_100	poison	silver	standard
badge_poison_gold	Gold Poison	gold	fruit_poison_500	poison	gold	standard
badge_poison_mythic	Mythic Poison	mythic	fruit_poison_2500	poison	mythic	animated
badge_golden_apple	Golden Apple	gold	fruit_apple_500	apple_gold	none	standard
badge_fruit_salad	Fruit Salad	common	fruit_salad	salad	none	standard
badge_toxic_slime	Toxic Slime	rare	fruit_toxic_appetite	slime	none	standard
badge_poison_shield	Poison Shield	rare	fruit_poison_proof	shield	none	standard
badge_apple_calendar	Apple Calendar	rare	fruit_apple_a_day	calendar	none	standard
badge_apple_basket	Apple Basket	gold	fruit_orchard_raider	basket	none	standard
badge_pure_apple	Pure Apple	rare	fruit_clean_eating	apple	none	standard
badge_natural_flyer	Natural Flyer	rare	fruit_no_boost	feather	none	standard
badge_questionable_fruit	Questionable Fruit	common	fruit_bad_decisions	poison	none	standard
badge_tiny_wings	First Flight	common	survival_30	wings1	bronze	standard
badge_silver_wings	Learning to Fly	rare	survival_60	wings2	silver	standard
badge_gravity	Gravity Defier	gold	survival_120	gravity	gold	standard
badge_eternal_wings	Sky Eternal	mythic	survival_300	wings3	mythic	animated
badge_falling_star	Falling Star	gold	survival_terminal_velocity	star_fall	none	standard
badge_scraped_paw	Scraped Paw	common	survival_close_call	paw	none	standard
badge_razor_edge	Razor Edge	rare	survival_living_on_edge	edge	none	standard
badge_balance	Balance	rare	survival_perfect_control	balance	none	standard
badge_empty_sky	Empty Sky	rare	survival_untouchable	empty	none	standard
badge_last_tap	Last Tap	rare	survival_one_more_tap	tap	none	standard
badge_coin_collector	Coin Collector	rare	coins_balance_500	coin	silver	standard
badge_golden_vault	Golden Vault	gold	coins_balance_2500	vault	gold	standard
badge_token_hunter	Token Hunter	gold	coins_token_hunter	coin_stack	none	standard
badge_shopping_bag	Shopping Bag	rare	coins_big_spender	bag	none	standard
badge_merchant	Merchant	rare	coins_shop_regular	tag	none	standard
badge_wardrobe	Wardrobe	rare	coins_fully_dressed	wardrobe	none	standard
badge_golden_hero	Golden Hero	rare	coins_golden_opportunity	star	none	standard
badge_dropped_coin	Dropped Coin	common	coins_shiny_distraction	coin_drop	none	standard
badge_first_crown	First Victory	gold	online_win_1	crown	none	standard
badge_survivor_crown	Last One Floating	rare	online_win_10	crown2	silver	standard

Badge ID	Display name	Rarity	Source achievement	Glyph	Tier	Flags
badge_champion_wings	Sky Champion	mythic	online_win_100	crown3	gold	animated
badge_photo_finish	Photo Finish	gold	online_photo_finish	photo	none	standard
badge_perfect_crown	Perfect Crown	gold	online_flawless	crown_clean	none	standard
badge_toxic_crown	Toxic Crown	gold	online_poisoned_champion	crown_toxic	none	standard
badge_comeback	Comeback	rare	online_comeback	comeback	none	standard
badge_crowded_sky	Crowded Sky	rare	online_crowded	crowd	none	standard
badge_rival	Rival	rare	online_rivalry	rival	none	standard
badge_friends	Friends	common	online_friendly	friends	none	standard
badge_beginner	Beginner	common	prog_getting_started	star_small	none	standard
badge_badge_case	Badge Case	rare	prog_badge_collector	case	none	standard
badge_achievement_hunter	Achievement Hunter	gold	prog_achievement_hunter	trophy	none	standard
badge_completionist	Completionist	mythic	prog_completionist	laurel	none	animated
badge_tap_counter	Tap Counter	rare	prog_dedicated_tapper	counter	silver	standard
badge_veteran	Veteran	gold	prog_tap_veteran	counter_gold	gold	standard
badge_never_quit	Never Quit	rare	prog_persistent	retry	none	standard
badge_weekly_wings	Weekly Wings	gold	prog_daily_flyer	calendar_wings	none	standard
badge_regret	Regret	common	secret_worth_it	regret	none	hidden
badge_floor	Floor Inspector	common	secret_floor_inspector	floor	none	hidden
badge_purple_skull	Purple Skull	gold	secret_maximum_poison	skull	none	hidden
badge_balance_fruit	Balance Fruit	rare	secret_perfectly_balanced	scales	none	hidden
badge_nine_lives	Nine Lives	gold	secret_nine_lives	cat	none	hidden
badge_greedy_hero	Greedy Hero	common	secret_too_greedy	coin_drop	none	hidden

The current JSON contains no Infinity-category achievements. Infinity achievement concepts in the design chapters and seed catalog remain planned/design-only until stable IDs, tracking fields, badge definitions, runtime progress, grant evidence, and migration behavior exist.

44. Current Implemented Challenge Catalog

Source basis: Workspace data/challenges.json. Daily and weekly pools are recorded here for reference and test coverage.

Cadence		Configured value			
Daily selection count		3			
Weekly selection count		3			
Daily completion bonus		5 Hero Coins			
Weekly completion bonus		Not configured separately			

Pool	ID	Name	Player-facing requirement	Metric / target	Reward
Daily	d_apples_10	Orchard Run	Collect ten Green Apples.	apples_collected x 10	3 coins
Daily	d_survive_60	One Minute	Survive for 60 seconds in a round.	rounds_survived_60 x 1	5 coins
Daily	d_rounds_5	Five Flights	Complete five rounds.	rounds_played x 5	3 coins
Daily	d_tokens_3	Gold Rush	Collect three Golden Hero Tokens.	hero_tokens_collected x 3	4 coins
Daily	d_taps_250	Busy Paws	Tap 250 times.	taps x 250	4 coins
Daily	d_no_fruit	Fasting	Finish a round without collecting fruit.	rounds_no_fruit x 1	4 coins
Daily	d_poison_3	Bad Diet	Consume three Poison Fruits.	poison_fruits_consumed x 3	3 coins
Daily	d_coins_20	Pocket Change	Earn 20 Coins through gameplay.	gameplay_coins_collected x 20	5 coins
Daily	d_online_1	Into the Fray	Play one online match.	online_matches_played x 1	5 coins; online
Daily	d_tokens_8	Token Hoarder	Collect 8 hero tokens.	hero_tokens_collected x 8	5 coins
Daily	d_cookies_5	Star Baker	Eat five Space Cookies.	space_cookies_eaten x 5	4 coins
Daily	d_cookies_12	Sweet Tooth	Eat twelve Space Cookies.	space_cookies_eaten x 12	6 coins
Weekly	w_survive_1200	Twenty Minutes Up	Survive for a combined 20 minutes.	total_survival_seconds x 1200	25 coins
Weekly	w_wins_10	Ten Crowns	Win ten online matches.	online_matches_won x 10	25 coins; online
Weekly	w_apples_100	Harvest	Collect 100 Green Apples.	apples_collected x 100	20 coins
Weekly	w_poison_25	Acquired Taste	Consume 25 Poison Fruits.	poison_fruits_consumed x 25	15 coins
Weekly	w_coins_100	Paid Work	Earn 100 Coins through gameplay.	gameplay_coins_collected x 100	20 coins
Weekly	w_dailies_5	Regular	Complete five daily challenges.	daily_completions x 5	20 coins
Weekly	w_rounds_50	Fifty Flights	Play 50 rounds.	rounds_played x 50	15 coins
Weekly	w_taps_5000	Relentless	Tap 5,000 times.	taps x 5000	25 coins
Weekly	w_cookies_40	The Whole Jar	Eat 40 Space Cookies.	space_cookies_eaten x 40	20 coins

FAIRNESS Challenge rotation must not create pay-to-win pressure, require an unavailable queue, punish players for accessibility settings, or silently replace earned progress. Online-only tasks must be clearly labeled before the player commits time.

45. Current Implemented Store and Equipment Catalog

Source basis: *Workspace data/store.json* plus the equipment-slot rules consolidated in this Bible.

Slot	ID	Name	Price	Color	Description	Bundle grants
skin	default	Velvet Hex	0 coins	#8f5fd6	Nine lives, allegedly. Sceptical about gravity.	-
skin	hex_toxic	Toxic Hex	100 coins	#7dff4a	Something is leaking. It does not seem worried.	-
skin	hex_golden	Golden Hex	100 coins	#ffd45e	Heavier than it looks. Falls exactly as fast.	-
skin	hex_arcade	Arcade Hex	100 coins	#31d6ff	Ships with scanlines nobody asked for.	-
skin	hex_frozen	Frozen Hex	100 coins	#9fe8ff	Rimmed over. Still not cool about it.	-
skin	hex_molten	Molten Hex	100 coins	#ff6a1f	The cracks glow. That is probably fine.	-
skin	hex_ghost	Ghost Hex	100 coins	#dfe6ff	Barely there. Still hits the floor.	-
skin	hex_corrupted	Corrupted Hex	100 coins	#ff2f6a	Some of these pixels are in the wrong place.	-
skin	hex_strawberry	Strawberry Slime	100 coins	#e8547f	Seeded. Sweet. Structurally questionable.	-
skin	hex_abyssal	Abyssal Hex	100 coins	#2fd8c8	Grew its own light. Down there you have to.	-
skin	hex_meadow	Meadow Hex	100 coins	#8fd66a	Flowers in the fur. Nothing has ever gone wrong.	-
skin	hex_first_contact	First Contact Hex	100 coins	#8fff6a	Being taken. Has made peace with it.	-
skin	hex_cosmic	Cosmic Hex	200 coins	#a88fff	A nebula moves through it, and it never moves the same way twice.	-
skin	hex_bundle	The Hex Bundle	1000 coins	#c9a8ff	Every variant, including Cosmic. Worth 1300 coins bought one by one.	hex_toxic, hex_golden, hex_arcade, hex_frozen, hex_molten, hex_ghost, hex_corrupted, hex_strawberry, hex_abyssal, hex_meadow, hex_first_contact, hex_cosmic
skin	hex_completionist	Prismatic Hex	earned only	#cfc4e8	Earned, not bought. Not in the bundle, not for sale, not ever.	-
trail	none	No Trail	0 coins	#7a819c	Clean flight.	-
trail	trail_spark	Sparks	35 coins	#ffd45e	Leaves a lick of gold behind each tap.	-
trail	trail_comet	Comet	70 coins	#6fd0ff	A cold streak across the sky.	-
trail	trail_bloom	Bloom	110 coins	#ff8fc7	Petals where you have been.	-

Slot	ID	Name	Price	Color	Description	Bundle grants
effect	none	No Effect	0 coins	#7a819c	A silent tap.	-
effect	fx_ring	Shockring	30 coins	#8fd8ff	A ring of air on every tap.	-
effect	fx_burst	Starburst	55 coins	#ffd45e	Little stars where you flapped.	-
effect	fx_ripple	Ripple	80 coins	#a0f0c8	The sky remembers each beat.	-

The shipped catalog currently exposes one skin slot, one trail slot, and one tap-effect slot. The source comment requires globally unique IDs, yet 'none' appears for both trail and effect, and a server map keyed only by item ID would collide. Treat this as a known implementation defect until the IDs are made unique or every lookup and persistence key is proven composite. Prismatic Hex is explicitly earned-only and must remain absent from direct-purchase and bundle grants.

IMPLEMENTATION BOUNDARY A current catalog record describes available content, not permission for the content to affect physics or outcomes. Ownership, equipped selection, ledger entries, acquisition source, refunds, and catalog version must persist independently.

46. Canonical Data Model Reference

Source basis: Cross-source synthesis of accounts, competitive state, Infinity state, economy, content ownership, integrity, and configuration requirements.

The names below are conceptual contracts, not mandated database table names. Every durable grant uses an idempotency key; every mutable record carries a schema/version field; server-authored competitive and economy state is never trusted from an unverified client payload.

Model	Required fields	Invariant
Shared user	user ID; provider links; display name; locale; age/consent state; account creation; sanctions; cross-save revision; settings reference	Identity links can merge, but product-specific progression remains separate.
Wallet ledger	transaction ID; user; currency; signed amount; reason; source product; source event; idempotency key; timestamp; balance after; reversal link	The ledger is authoritative; cached balances are derived and reconciled.
Inventory ownership	user; slot/type; item ID; acquired at; source; product origin; paid/earned; permanent flag; content version	Infinity-exclusive acquisition source remains visible and cannot be rewritten as purchased.
Equipment	user; product; slot; equipped item; updated at; revision	Only owned and product-eligible items may equip; one item per slot unless a later rule says otherwise.
Competitive profile	rating; RD; volatility; placement count; tier; division; peak/final rank; season; wins/losses; streak; human/bot split	Visible rank is presentation; rating is the matchmaking signal.
Performance profile	survival median; recent windows; input quality; disconnect pattern; platform/network indicators	Supports match quality and integrity; never exposes sensitive risk scoring to opponents.
Match	match ID; ruleset/config version; participants; bot flags; seed; start/end; placements; survival; pickups; disconnects; evidence; rating delta	Server or authoritative host validates results and finalizes once.
Bot definition	personality; tier target; reaction distribution; miss rate; impulse policy; pursuit bias; loadout; disclosure flag	Bots follow the same physics and item rules; no hidden immunity or impossible input.
Season	season ID; start/end; ruleset; rating compression; reward table; eligibility; archival state	Past peak/final ranks and earned rewards remain auditable.
Infinity save	world/biome/zone; distance; altitude/depth records; tap totals/remainder; Journal sets; interactions; rare/pity state; pending grants	Local-first progress syncs safely; merge cannot duplicate currency or erase set-valued discoveries.
Journal entry	entry type/ID; discovered state; progress; first source/location; exact variant; timestamps; hidden-hint state	Supports partial behavior counts without revealing every secret.
Achievement progress	achievement ID/version; current value; completed; claimed; evidence scope; awarded transaction	Retroactive totals and online evidence are distinct.
Challenge assignment	rotation ID; challenge ID/version; assigned/expiry; progress; completion; claim; authoritative scope	Expiry never removes already granted rewards.
Catalog item	item ID; slot; price; currency; availability; grants; restrictions; art/audio refs; content version	Store visibility is independent from ownership and equipment.
Interaction definition	target; tap thresholds; cooldown; conditions; weighted variants; chain outputs; Journal effect; rare eligibility	Drives Infinity depth without hard-coding each object.
Rare discovery state	category; eligibility; attempts/taps; pity; last check; last success; pending result; idempotency key	Never expose exact hidden thresholds in the client.
Integrity case	signals; match/account references; confidence; automated action; reviewer; appeal; resolution; retention class	Enforcement must be explainable, proportionate, and auditable.

Model	Required fields	Invariant
Remote configuration	config ID/version; audience; rollout; start/end; checksum; rollback; author/approval	A match pins one immutable ruleset version from start through result.

Cross-Device Merge Rules

- Set-like discoveries merge by union, preserving earliest known source and timestamp when available.
- Monotonic records such as highest altitude use max only when both records belong to the same measurement version and are not flagged invalid.
- Wallet state merges by replaying authoritative ledger transactions, never by choosing the larger cached balance.
- Tap remainder toward the next 1,000-tap award merges from transaction/evidence history; adding two remainders can duplicate credit and is forbidden.
- Exclusive grants reconcile by item ownership ID and grant idempotency key. Pending grants retry until acknowledged, not until duplicated.
- Conflicting current locations use a documented policy such as most recent valid checkpoint, while permanent discoveries remain unioned.
- Competitive rating accepts only authoritative finalized matches. Offline clients cannot advance rating or backfill ranked results.

47. Remote Configuration and Analytics Event Reference

Source basis: Cross-source implementation and observability requirements. Names are canonical semantic events; transport-specific naming may differ if mapped one-to-one.

Domain	Configuration	Safety rule
Competitive physics	gravity curve; tap impulse; velocity caps; collision tolerances; valid-tap timing	Pinned per ruleset/match; signed; rollback tested.
Pickups and hazards	spawn cadence; positions; weights; durations; stacking; cleanse; despawn	Same authoritative values for humans and bots.
Matchmaking	rating weight; survival weight; network weight; search bands; expansion timers; bot-fill thresholds	Never change an already formed match.
Ranked	tier thresholds; divisions; placement count; season length; compression; rewards	Archive the exact version with season results.
Day/night	120-second phase timing; colors; anchors; cloud/stars/celestial speeds; quality tier	Independent from gravity; deterministic from shared match time.
Infinity movement	impulse profiles; drag; speed limits; transition thresholds; camera smoothing	Cannot introduce loss or punishment.
Infinity content	zone sequences; interaction thresholds; rare tables; pity; unlock mapping	Content version stored with discoveries and grants.
Economy	prices; rewards; challenge selection; caps; offers	Ledger reason/version recorded; paid and earned paths remain distinguishable.
Cosmetics	particle caps; lifetimes; remote-player density; quality tiers; accessibility overrides	No config may alter competitive outcomes.
Audio	bus levels; variation; cooldowns; ducking; compression/streaming	Accessibility overrides win over event tuning.

Event family	Minimum properties	Purpose / privacy constraint
app_session	product; build; device tier; locale; consent; start/end reason	Retention and stability by product.
onboarding_step	step; entered/completed/skipped; time; failure	Tutorial clarity without coercion.
tap_validated	product; context; valid/invalid reason; sampled timing bucket	Input quality and false rejection; do not log raw touch traces unnecessarily.
competitive_match	queue; ruleset; humans/bots; survival; placement; disconnect; result	Match health, fairness, population, and bot prevalence.
rating_change	season; before/after; RD; expected result; placement flag	Rating correctness and volatility.
pickup_event	item; spawned/collected/expired; target; outcome	Spawn balance and item comprehension.
infinity_zone	world; biome; zone; entered/completed/looped; duration	Pacing, depth, and transition health.
interaction_event	target; stage; condition; chain; Journal result	Which living-world reactions are found and repeated.
discovery_event	type; rarity; eligible attempts; pity band; source; grant result	Rare-system droughts and grant integrity without exposing secret criteria publicly.
economy_transaction	currency; amount; reason; source; item; ledger ID	Source/sink balance, fraud, and reconciliation.
catalog_action	view; preview; purchase; equip; fail reason; product origin	Store clarity, conversion, and mistaken purchase detection.
achievement_challenge	assigned/progress/completed/claimed; ID/version; evidence scope	Progress correctness and unreachable content.

Event family	Minimum properties	Purpose / privacy constraint
accessibility_setting	setting changed; anonymous cohort only	Discover defaults and usability gaps; never infer disability.
performance_sample	device tier; FPS; memory; load; thermal; battery; network	Quality-tier tuning and regression detection.
integrity_signal	coarse signal; confidence; action; appeal outcome	Measure false positives with strict access and retention.

Canonical Suggested System Event Names

These stable semantic names come directly from the detailed Game Bible event appendix. Implementations may add a namespace or version suffix, but dashboards and tests should preserve a documented one-to-one mapping.

Event group	Canonical event names
Input and run	tap_registered; player_impulse_applied; ground_contact; run_started; run_ended; close_call; personal_best
Match	match_started; match_completed; disconnect; reconnect
Pickups	pickup_spawned; pickup_collected; green_apple_collected; poison_fruit_collected; hero_token_collected
Progression	achievement_progress; achievement_earned; reward_claimed; badge_equipped; challenge_progress; challenge_completed
Ranked and season	rating_updated; tier_changed; season_reward_granted
Infinity movement	infinity_tap_milestone; biome_entered; zone_transitioned; boost_started; boost_ended
Infinity discovery	environment_reacted; interaction_chain_advanced; animal_behavior_discovered; hero_encountered; secret_discovered; exclusive_unlocked
Commerce/entitlement	purchase_validated; ad_free_started; ad_free_expired
Integrity	integrity_flag_created

48. Cross-System Definition of Done and Release Gates

Source basis: Consolidated developer acceptance criteria across all supplied bibles, expanded into product-level release evidence.

Gate	Pass condition	Required evidence
Competitive input/physics	Repeated taps produce consistent upward resistance; gravity escalation follows pinned config; collisions and ground loss are deterministic; capped inputs do not feel arbitrary; pause/background behavior is defined.	Automated deterministic replays plus frame-rate/device matrix and extended manual feel test.
Competitive fairness	No owned cosmetic, purchase, accessibility option, device quality tier, or remote effect changes collision, visibility of required signals, item odds, or rating outcome.	Golden-match comparisons across clean and fully equipped accounts.
Multiplayer authority	Two or more clients agree on match seed, phase, pickups, elimination order, result, and grant; reconnect and duplicate result submission are idempotent.	Two-device tests under latency, loss, suspend/resume, reconnect, and retry.
Ranked	Glicko values update once; placements complete at configured count; visible tier maps correctly; season compression/rewards archive; division count is resolved before shipping ladder UI.	Reference calculations, migration tests, season rollover rehearsal, abuse cases.
Bots	Bots use the same physics and pickups; declared personalities produce measurable behavior; bot identity is disclosed; impossible reactions and hidden protection are absent.	Seeded simulation, behavior distributions, human/bot result split, spectator review.
Day/night	All clients derive the same 120-second presentation from shared match time; phase boundaries are smooth; brightness never encodes gravity; required objects remain readable at darkest phase.	Two-device time-offset capture, color-contrast checks, tier comparison, reduced-motion/flash review.
Infinity contract	No loss, punishment, urgency, or compulsory monetization; movement is satisfying without rewards; stop/resume preserves progress; one deep biome meets the vertical-slice gate.	Long idle/fidget sessions, interruption matrix, new-player observation, no-network validation.
Infinity taps/currency	Exactly one Hero Coin is granted per 1,000 valid lifetime taps; remainders persist; reconnect, merge, retry, and reinstall do not duplicate or erase credit.	Boundary tests at 999/1000/1001, multi-device merge, crash between grant and acknowledgement.
Infinity discovery	Interaction stages, context, cooldown, chains, Journal progress, rare eligibility, pity, and pending grants persist; animals are never harmed; secret criteria do not leak.	Content-driven interaction harness, save migration, deterministic rare simulations, copy/log audit.
Exclusive rewards	Every designated Infinity-exclusive item is earnable in Infinity, visible with source, and absent from all shops, bundles, passes, ads, and paid grants.	Automated catalog invariant across every offer and server grant path.
Economy/inventory	Ledger balances reconcile; ownership is permanent unless explicitly reversible; duplicate grants collapse; refunds/reversals are traceable; equip validates slot and ownership.	Transaction property tests, offline/retry cases, migration and rollback rehearsal.
Achievements/challenges	All enabled records can progress; retroactive and online-only evidence are correct; hidden criteria remain hidden; rotation never assigns impossible content; grants occur once.	Data-driven completion suite covering every catalog row and version migration.
Cosmetics/effects	Tap effects, trails, particles, skins, badges, frames, and elimination effects remain readable, pooled, capped, accessible, and mechanically inert.	Performance stress scene, remote-player density test, accessibility toggle audit, golden physics comparison.
Audio	Tap, pickup, danger, result, and menu cues remain distinct; repeated sounds vary without fatigue; streaming/compression works; mute and per-bus controls persist.	Device/headphone/speaker matrix, rapid-tap fatigue session, interruption and Bluetooth changes.
Accessibility/UI	Text scales/reflows; touch targets, contrast, reduced motion, flash, haptic, audio, remote effects, and color-independent cues work; destructive/paid actions confirm clearly.	Screen-reader/navigation review, scale matrix, one-handed use, color-vision simulation, comfort sessions.

Gate	Pass condition	Required evidence
Performance/mobile	Target device tiers sustain agreed FPS, memory, load time, battery, thermal, and network budgets; quality reductions preserve gameplay information.	Cold/warm launch, 30-minute soak, thermal/battery sampling, low-memory recovery, poor-network profile.
Security/moderation	Server rejects impossible results and duplicate grants; sanctions and appeals are recorded; display names/reactions are bounded; privacy retention is documented.	Threat-model cases, tampered-client tests, rate limits, false-positive audit, moderation workflow rehearsal.
Analytics/privacy	Required semantic events arrive once with pinned versions; consent and retention rules are honored; dashboards distinguish products, humans/bots, configs, and acquisition source.	Schema validation, sampled end-to-end trace, opt-out/deletion test, access-control review.

SHIP RULE A milestone is complete only when its acceptance evidence exists. A feature being visible in a build is not the same as being production-ready.

49. Canonical Open-Decision Register

Source basis: Unresolved choices collected across all supplied sources. Owners and dates belong in the production tracker; the Bible records what must be decided and why.

Decision	Question to resolve	Downstream impact
Product packaging	Whether Competitive TAM and Infinity Taps ship as separate store apps, one shell with two entries, or another arrangement while remaining separate player-facing experiences.	Account linking, store policy, acquisition, downloads, reviews, release cadence.
Shared ownership policy	Which cosmetics earned or purchased in one product may equip in the other, and how unavailable assets display.	Collection clarity, download size, entitlement support.
Rank divisions	Resolve the conflict between five divisions (V-I) and three divisions (III-I) inside each tier.	Rank UI, thresholds, rewards, migrations, season messaging.
Rank thresholds	Approve rating-to-tier thresholds and whether visible progress is direct rating, stars, or a protected presentation layer.	Player trust and volatility.
Season rewards	Exact rewards, eligibility, peak versus final basis, inactivity handling, and distribution timing.	Competitive motivation and support burden.
Placement/bot rules	When placements may include bots, minimum human count, and how bot-heavy results affect rating.	Queue health versus legitimacy.
Queue expansion	Final timing/bands by region, mode, skill, and population, including maximum acceptable latency.	Wait time, fairness, network quality.
Disconnect outcome	Reconnect window, temporary AI takeover if any, rating consequences, and abuse detection.	Mobile reliability and competitive integrity.
Gravity curve	Final escalation formula, soft/hard caps, round-length target, and interaction with item timing.	Core skill arc and match duration.
Day/night rollout	Whether the full 120-second system launches with Competitive MVP or follows after core multiplayer stability.	Scope, visual identity, cross-device synchronization.
Competitive item set	Launch pickups/hazards, targeting, spawn weights, cleanse rules, and stacking matrix.	Readability, comeback potential, bot parity.
Infinity launch scope	One-biome vertical slice versus broader initial content, and the exact first biome/zone sequence.	Depth, production cost, retention.
Infinity movement profiles	Final impulse/drag/camera values for land, underwater, sky, space, and other environments.	Feel and accessibility.
Zone progression	Exact approximately-two-minute pacing, branching/loop policy, checkpoint behavior, and transition presentation.	Fidget continuity and exploration agency.
Rare system	Check interval, base odds, pity curve, drought reset, category pools, and disclosure policy.	Wonder, fairness, simulation risk.
Hero acquisition	Which heroes/variants are found, earned, purchased, or hybrid; duplicate handling; two-step encounter conversion details.	Collection ethics and revenue.
Infinity exclusives	First reward set, unlock thresholds, cross-product equipping, and permanent source labels.	Mode identity and non-purchase promise.
Currency caps	Daily/weekly reward caps, AFK/automation protection, and treatment of offline Infinity taps.	Inflation, accessibility, fraud.
Economy prices	Approve price bands, earning velocity, bundles, refunds, regional paid pricing, and supporter offers.	Ethical monetization and progression pace.
Advertising	Whether ads ship at all; if so, exact opt-in placements and non-punitive reward rules.	Trust and platform compliance.
Effect depth	Which effects receive accumulation, reactions, or rare events at launch versus simple presentation only.	Content cost, performance, discovery coherence.

Decision	Question to resolve	Downstream impact
Character roster	First playable/collectible heroes beyond Velvet Hex, variant rarity, lore depth, and product availability.	Brand breadth and production capacity.
Narrative voice	Final naming, capitalization, humor limits, mystery density, and localization voice guide.	Brand consistency and translation.
Audio motif	Approve the shared TAM melodic/rhythmic motif, instrumentation per product, and adaptive layer budget.	Cross-product recognition and file size.
Account model	Guest-to-account conversion, provider set, age/consent handling, deletion, merge conflicts, and recovery.	Cross-save, compliance, support.
Multiplayer architecture	Final authoritative topology, region strategy, session service, relay/host fallback, and replay evidence.	Cheat resistance, latency, cost.
Moderation scope	Allowed social features, name/reaction policy, report/appeal flow, and retention.	Safety and operational burden.
Analytics stack	Event transport, warehouse, consent implementation, retention, dashboards, and ownership.	Decision quality and privacy.
Device floor	Minimum Android/API/device tier, FPS targets, memory/load/battery/thermal budgets, and asset packs.	Audience reach and quality.
Accessibility defaults	Default flash, shake, haptics, particle density, remote cosmetics, text scale, and color support.	Comfort and inclusion.
Live-ops cadence	Season/event cadence, content capacity, reruns, permanent ownership, and rollback expectations.	Sustainable production.
Legacy concepts	Which reserved modes, items, effects, or product structures are retired permanently versus held for later validation.	Scope control and documentation clarity.

Decision Record Template

Field	Required content
Decision ID / title	Stable reference used by specs, tickets, tests, and release notes.
Owner / approvers	One accountable owner plus required product, engineering, art, audio, legal, or operations approval.
Status / date	Proposed, approved, superseded, or retired; effective date and target release.
Context	Player problem, evidence, constraints, affected products and systems.
Options	Alternatives considered, including cost, risk, accessibility, fairness, performance, and live-ops impact.
Decision / rationale	Exact approved behavior and why it best protects the product contracts.
Configuration / migration	Tunable fields, default values, schema/content version, backward compatibility, rollback.
Acceptance evidence	Tests, playtests, metrics, review artifacts, and release gate.
Supersedes / links	Older Bible language, code spec, task, asset brief, analytics dashboard, and support notes.

50. Source Concordance and Authority Map

Source basis: Traceability map for the nine supplied source files. Page counts and hashes appear in the front matter.

Source	Principal contribution	Authority treatment
TAM Game Design Bible v1.0 (DOCX)	Primary canonical narrative for governance, two-product architecture, all system chapters, roadmap, decisions, terminology, and acceptance framing.	Newest explicit decisions control unless a newer feature-specific authority says otherwise.
Competitive Dynamic Day-Night and Gravity Presentation Bible	Exact 120-second competitive environment cycle, independent gravity presentation, controller architecture, sync, visual anchors, quality tiers, accessibility, phased delivery, acceptance.	Feature-specific authority for competitive sky/time presentation.
Original Expanded Bible	Deep tap effects, character particles, price bands, Infinity worlds/interactions/exclusives, rare/pity concepts, ranked/matchmaking/bot details, monetization ideas.	Richest backlog source; proposals remain candidate or tuning unless promoted.
Game Design Bible (1) / Game Design Bible	Consolidated product and implementation snapshot, developer acceptance, current-vs-planned distinctions.	These two supplied PDFs are byte-identical and count as one unique authority.
TAM Game Bible (DOCX)	Broad product, content, art, audio, technology, economy, and roadmap synthesis with substantial nuance.	Supporting comprehensive source; newer explicit decisions prevail.
Game Bible v1.0 (DOCX)	Detailed mechanics, system behavior, implementation snapshot, handoff detail, bot/ranked/Infinity/product notes.	Supporting engineering/design authority.
Game Design Bible v1.0 (PDF)	Design principles, mechanics, production phases, acceptance framing, and scope controls.	Supporting design/production authority.
TAM Design Bible v1.0 (PDF)	Condensed high-level vision, product framing, identity, and prioritized direction.	Supporting executive/design summary.

Topic-to-Source Trace

Topic	Primary source basis	Master-Bible location
Product architecture and brand	TAM Game Design Bible; TAM Game Bible; condensed design bibles	Chapters 1-6
Competitive core, physics, pickups	All consolidated bibles; implementation snapshots	Chapters 7-12
Ranked, Glicko, matchmaking, bots	Original Expanded Bible; Game Bible v1.0; consolidated design bibles	Competitive implementation reference
Competitive day/night	Dedicated Dynamic Day-Night and Gravity Presentation Bible	Dedicated day/night specification
Infinity contract and movement	Original Expanded Bible; TAM Game Design Bible; TAM Game Bible	Chapters 13-16 and Infinity implementation reference
Living-world interaction and Journal	Original Expanded Bible; consolidated design bibles	Infinity implementation reference
Heroes, variants, collection	TAM Game Design Bible; TAM Game Bible; Original Expanded Bible	Hero/collection chapters and current store appendix
Effects and particles	Original Expanded Bible; workspace store catalog; consolidated design bibles	Expanded effect catalog and current store appendix
Economy, achievements, challenges	Original Expanded Bible; consolidated bibles; workspace JSON catalogs	Progression chapters and implemented catalog appendices
Audio, UI, accessibility	TAM Game Bible; Game Bible v1.0; design bibles	Audio/UI/mobile-quality chapters and release gates
Backend, security, analytics	Game Bible v1.0; TAM Game Design Bible; consolidated PDFs	Technology chapters and data/event appendices
Roadmap, live ops, risks	All consolidated bibles	Delivery chapters, decision register, release gates

MAINTENANCE When a new decision changes this Bible, update the relevant chapter, status, open-decision row, acceptance gate, configuration/data contract, source note, and version history together. Do not append a contradictory note and leave the old rule appearing equally valid.